# Tree mortality and ecosystem functional properties

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*Coauthor notes (not part of the manuscript). Section order follows Nature Ecology & Evolution: Abstract, Main, Results, Discussion, Methods, Data availability, Code availability, References, Acknowledgements, Funding, Author information, Extended Data / Supplementary Information.*

**Main findings** (AutoGluon, deadtrees v5, 101 sites, 361 site-years)

1. **Knowing about tree mortality makes carbon uptake more predictable, especially in the years when trees died.** Adding mortality lowers the error (MAE) of $\mathrm{NEP}_{max}$ by 11–12 % over all site-years and by 29–32 % in high-mortality years; WUE improves mainly where mortality happened (−17 to −22 % in high-mortality years); $\mathrm{GPP}_{sat}$ improves less (−4 to −12 %) and $\mathrm{ET}_{max}$ not at all (Table 2).
2. **Without mortality, models over-estimate carbon uptake in ecosystems with high mortality; mortality corrects this bias.** At >30 % relative mortality the model without mortality over-predicts $\mathrm{NEP}_{max}$ by about 4 and $\mathrm{GPP}_{sat}$ by about 4.3 $\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ and WUE by about 0.8 $\mathrm{g\,C\,mm^{-1}}$. The bias grows with mortality; adding mortality reduces it by about two-thirds, and the correction is significant — for all three learners and robust to resampling whole sites, leaving out any single site and removing the most extreme sites (Fig. 5).
3. **The effect follows the amount of mortality, not the ecosystem type.** Mortality pushes predicted $\mathrm{GPP}_{sat}$, $\mathrm{NEP}_{max}$ and WUE down at high-mortality sites (86–91 % of them) and slightly up at low-mortality sites (2–10 %); the direction tracks relative mortality closely ($\rho = -0.78$ to $-0.83$) and does not depend on tree cover or vegetation class. Relative mortality alone carries most of the information (Figs. 2–4).

*In one sentence:* tree mortality mapped from remote sensing makes ecosystem carbon uptake more predictable — without it, models over-estimate the productivity of woody ecosystems with high mortality, and the correction scales with the amount of mortality regardless of ecosystem type.

**What changed in this version**

- **Dataset:** deadtrees v5 mortality maps (delivered 2026-10-06).
- **Main learner: AutoGluon** (AutoML ensemble). XGBoost and LightGBM (Optuna-tuned) are in the Appendix as a check.
- **Main test:** paired bootstrap of the MAE at the **site-year level** (every year is an observation, whole sites resampled; Table 2). Site-level tests and the per-site figures are in the Appendix.
- Only **relative mortality** is shown in the main text; the other mortality metrics are in the Appendix.

# Abstract

*Draft for Nature Ecology & Evolution: unreferenced, ≤200 words.*

Tree mortality is increasing in forests worldwide, yet how it changes ecosystem functioning across sites remains poorly quantified. Here we combine eddy-covariance measurements from 101 sites with at least 30 % tree cover (361 site-years; forests, woody savannas, shrublands and wetlands) with remote-sensing maps of tree mortality and machine learning, and show that knowing about mortality makes their carbon uptake markedly more predictable. Adding mortality to climate and plant traits reduced the prediction error of maximum net ecosystem productivity by 11–12 % overall and by about 30 % in years with high mortality, and improved water-use efficiency mainly where mortality occurred; maximum evapotranspiration did not improve. Without mortality, models overestimated carbon uptake and water-use efficiency in high-mortality years, a bias that grew with the amount of mortality and that mortality information reduced by about two-thirds. The direction of the effect followed the amount of mortality rather than vegetation type or tree cover, and the results held across three learning algorithms and when the climate variables used to calculate each property were left out. Mortality maps can thus correct a systematic over-estimate of carbon uptake in woody ecosystems with high mortality.


# Main

- Terrestrial ecosystems regulate global **carbon, water, and energy balances**; forest functioning is closely linked to vegetation structure and functional traits.
- Increasing **drought, heat, and biotic disturbances** are causing widespread tree mortality, with potentially large consequences for ecosystem functioning.
- Mortality alters **canopy structure and leaf area**, affecting carbon uptake, evapotranspiration, and energy partitioning, with cascading effects on soils, nutrients, and vegetation recovery.
- The magnitude and trajectory of these impacts likely depend on **vegetation type and functional trait composition**.
- However, quantitative understanding of mortality impacts across ecosystems remains limited due to **spatial and temporal constraints of field observations** and limited integration of mortality and trait data.
- Advances in **remote sensing, machine learning, citizen science, and trait databases** now enable large-scale assessments of mortality and ecosystem responses.
- This creates an opportunity to quantify how mortality affects ecosystem functioning and identify **vegetation types that are more sensitive or resilient**.

# Results

### Figure 1: Sites and EFP distributions

- **101 sites, 361 site-years**, 2020–2025, tree cover $\geq$ 30 %: ENF 49, DBF 30, MF 7, WET 6, EBF 3, CSH 3, OSH 2, WSA 1. Median 4 years per site; 17 sites have a single year.
- Quality control: CN-Sb1 and CN-Sb2 (two Saihanba plantations with extreme WUE) and one negative-WUE site-year are excluded.

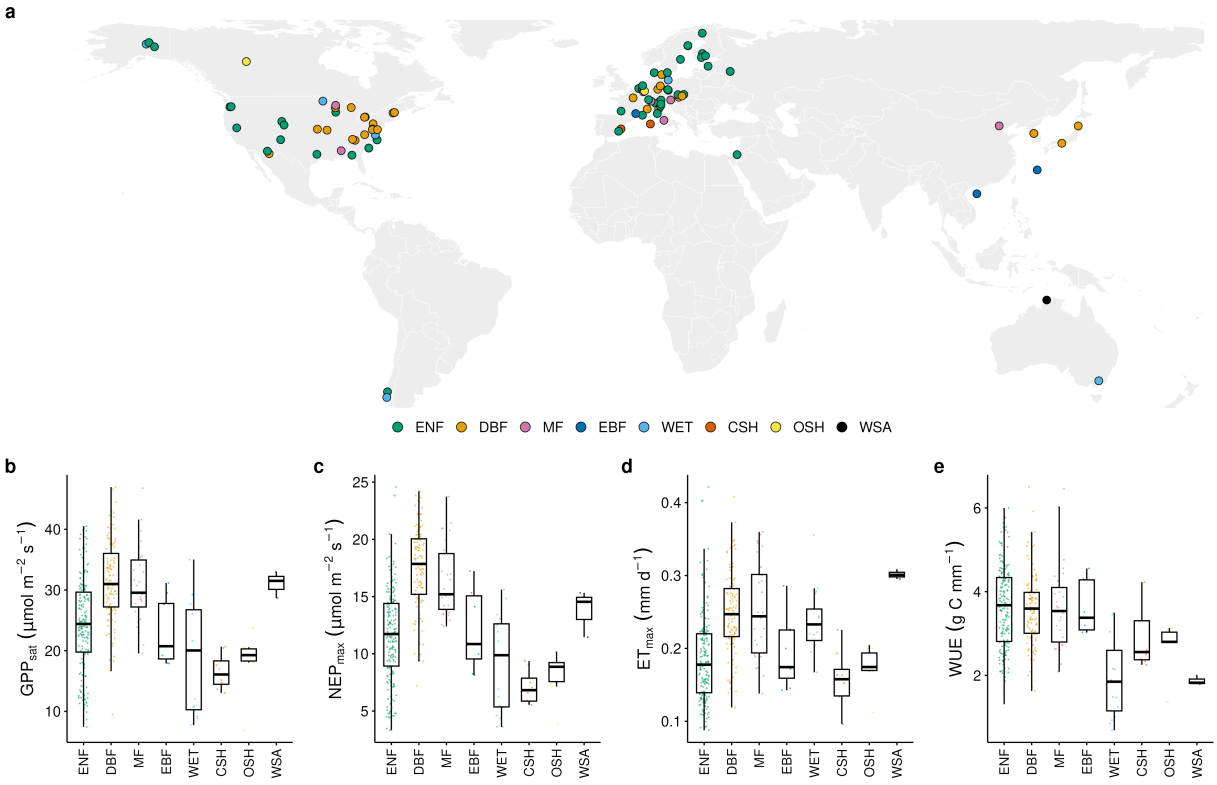



**Fig. 1** — sites and EFP distributions.

**Model skill (AutoGluon)**

- Repeated leave-one-site-out CV, $R^2$ of climate + traits (C+T) and climate + traits + mortality (C+T+M): adding mortality raises the mean $R^2$ from **0.349 to 0.402** (12 months) and from **0.351 to 0.408** (24 months) (Table 1).
- Largest gain for NEP$_{max}$ (+0.11), then WUE (+0.06) and GPP$_{sat}$ (+0.04–0.05); ET$_{max}$ unchanged.
- AutoGluon is slightly better than XGBoost and LightGBM on average (best in 7 of 16 tasks; Table 3).

Table 1: AutoGluon skill under repeated leave-one-site-out CV ($R^2$), without (C+T) and with mortality (C+T+M).

| | 12 months | | | 24 months | | |
|---|---|---|---|---|---|---|
| EFP | C+T | C+T+M | Δ | C+T | C+T+M | Δ |
| $GPP_{sat}$ | 0.330 | 0.367 | +0.037 | 0.317 | 0.370 | +0.053 |
| $NEP_{max}$ | 0.391 | 0.503 | +0.112 | 0.394 | 0.502 | +0.108 |
| $ET_{max}$ | 0.423 | 0.425 | +0.002 | 0.429 | 0.433 | +0.004 |
| WUE | 0.252 | 0.312 | +0.060 | 0.264 | 0.326 | +0.062 |
| Mean | 0.349 | 0.402 | +0.053 | 0.351 | 0.408 | +0.057 |

**Table 2: Does mortality reduce the prediction error?** (MAE, site-year level, paired bootstrap)

- Every site-year is an observation, so years without mortality do not dilute the years with mortality. Years of one site are not independent, so whole sites are resampled (bootstrap) and swapped (paired permutation).

- **NEP$_{max}$: yes, robustly.** MAE −11.8 / −10.8 % over all site-years (12 / 24 months), −17.5 to −22.6 % at High sites and −29 **to −32 % in high-mortality years**; significant in 7 of 8 cells.
- **WUE: only where mortality happened.** −4 to −5 % over all site-years (n.s.), but −11 to −15 % at High sites and −17 to −22 % in high-mortality years, all significant.
- **GPP$_{sat}$: smaller** (−4 to −12 %), significant over all site-years at 24 months and in the Mid + High years; at High sites and in High years the intervals are wide (few site-years).
- **ET$_{max}$: small** (−1 to −11 %), significant only in the Mid + High years at 24 months.
- **The effect is largest in the years when mortality happened**: for every EFP the error reduction in High years is at least twice the one over all site-years (NEP$_{max}$ about 2.7 times).
- The bootstrap intervals and the permutation tests agree: every significant cell has its 95 % interval entirely below zero.
- Robustness (Appendix): site-level tests (per-site MAE; log-ratio permutation and site-level bootstrap), RMSE instead of MAE, XGBoost and LightGBM, sites weighted by their number of years, one-year sites dropped. NEP$_{max}$ improves in all of them.

Table 2: Paired bootstrap of the change in MAE when mortality is added (C+T → C+T+M), AutoGluon. Cells: % change of the error, 100 (MAE$_{\mathrm{C+T+M}}$ / MAE$_{\mathrm{C+T}}$ − 1), with the 95 % bootstrap interval in brackets. *Site-year level*: every site-year is an observation (pooled MAE; no averaging over years). Years of one site are not independent, so whole sites are resampled (10 000 bootstrap draws) and the C+T / C+T+M labels are swapped per site for the paired permutation test (10 000 draws, one-sided; stars, Benjamini–Hochberg corrected across all three learners: $^{*}\,q < 0.05$, $^{**}\,q < 0.01$, $^{***}\,q < 0.001$). High sites: site maximum of relative mortality > 24.0 %; the last two columns keep only the site-years whose own relative mortality exceeds 10.6 % (Mid + High) or 24.0 % (High). Column heads: sites / site-years. 101 sites, 361 site-years, deadtrees v5.

| | | All sites (101 / 361) | High sites (22 / 80) | Mid + High years (60 / 160) | High years (22 / 40) |
|---|---|---|---|---|---|
| AutoGluon 12 months | $\mathrm{GPP_{sat}}$ | −4.1 [−8.0, −0.2] | −7.2 [−18.1, +2.9] | −6.7 [−12.8, −0.7]$^{*}$ | −12.0 [−29.5, +4.9] |
| | $\mathrm{NEP_{max}}$ | −11.8 [−17.0, −6.3]$^{**}$ | −22.6 [−33.6, −11.1]$^{**}$ | −12.5 [−20.1, −4.9]$^{*}$ | −31.6 [−43.2, −16.0]$^{*}$ |
| | $\mathrm{ET_{max}}$ | −1.3 [−4.2, +1.3] | −2.9 [−8.4, +2.1] | −3.9 [−7.7, +0.0] | −9.7 [−20.6, +2.7] |
| | WUE | −4.3 [−8.1, −0.2] | −10.9 [−17.8, −4.3]$^{*}$ | −7.2 [−12.2, −1.8]$^{*}$ | −16.9 [−25.2, −5.1]$^{*}$ |
| AutoGluon 24 months | $\mathrm{GPP_{sat}}$ | −5.7 [−9.5, −1.7]$^{*}$ | −7.1 [−17.2, +3.6] | −8.3 [−14.2, −2.6]$^{*}$ | −11.1 [−28.0, +8.1] |
| | $\mathrm{NEP_{max}}$ | −10.8 [−16.0, −5.1]$^{**}$ | −17.5 [−30.7, −2.1] | −11.6 [−19.8, −2.7]$^{*}$ | −29.2 [−42.9, −8.6]$^{*}$ |
| | $\mathrm{ET_{max}}$ | −2.0 [−5.0, +0.9] | −5.3 [−13.1, +1.2] | −5.4 [−9.6, −1.3]$^{*}$ | −11.4 [−23.7, +3.2] |
| | WUE | −4.7 [−9.1, −0.1] | −14.8 [−23.9, −6.4]$^{*}$ | −8.2 [−14.5, −1.5]$^{*}$ | −21.8 [−34.5, −8.2]$^{*}$ |

**Figure 2: Direction of the mortality effect at each site** (signed SHAP, relative mortality, 24 months)

- C+T+M model, per site the **net signed SHAP of the mortality block** (sum over the mortality predictors, averaged over the site's years, response units). Sites sorted by their maximum **relative mortality** (mean of the five buffers), highest on the left; dashed lines = Low / Mid / High breaks (10.6 % and 24.0 %). Below zero = the model predicts a *lower* EFP at that site because of its mortality values.
- **The sign follows the mortality class.** Share of sites with a negative contribution, Low / Mid / High: GPP$_{sat}$ 7 / 66 / 86 %; NEP$_{max}$ 10 / 68 / 91 %; WUE 2 / 58 / 86 %; ET$_{max}$ 24 / 50 / 77 %.
- **Size:** median −2.1 (GPP$_{sat}$) and −1.5 (NEP$_{max}$) $\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ at High sites, +1.4 and +1.2 at Low sites; ET$_{max}$ contributions are tiny ($\pm 0.005\,\mathrm{mm\,d^{-1}}$).

- Overall importance (share of mean |SHAP|, C+T+M): mortality 24 % ($GPP_{sat}$), 28 % ($NEP_{max}$), 16 % ($ET_{max}$), 23 % (WUE); climate 52–66 %; traits 11–20 %.
- The version with site names, the 12-month window and XGBoost / LightGBM are in the Appendix.

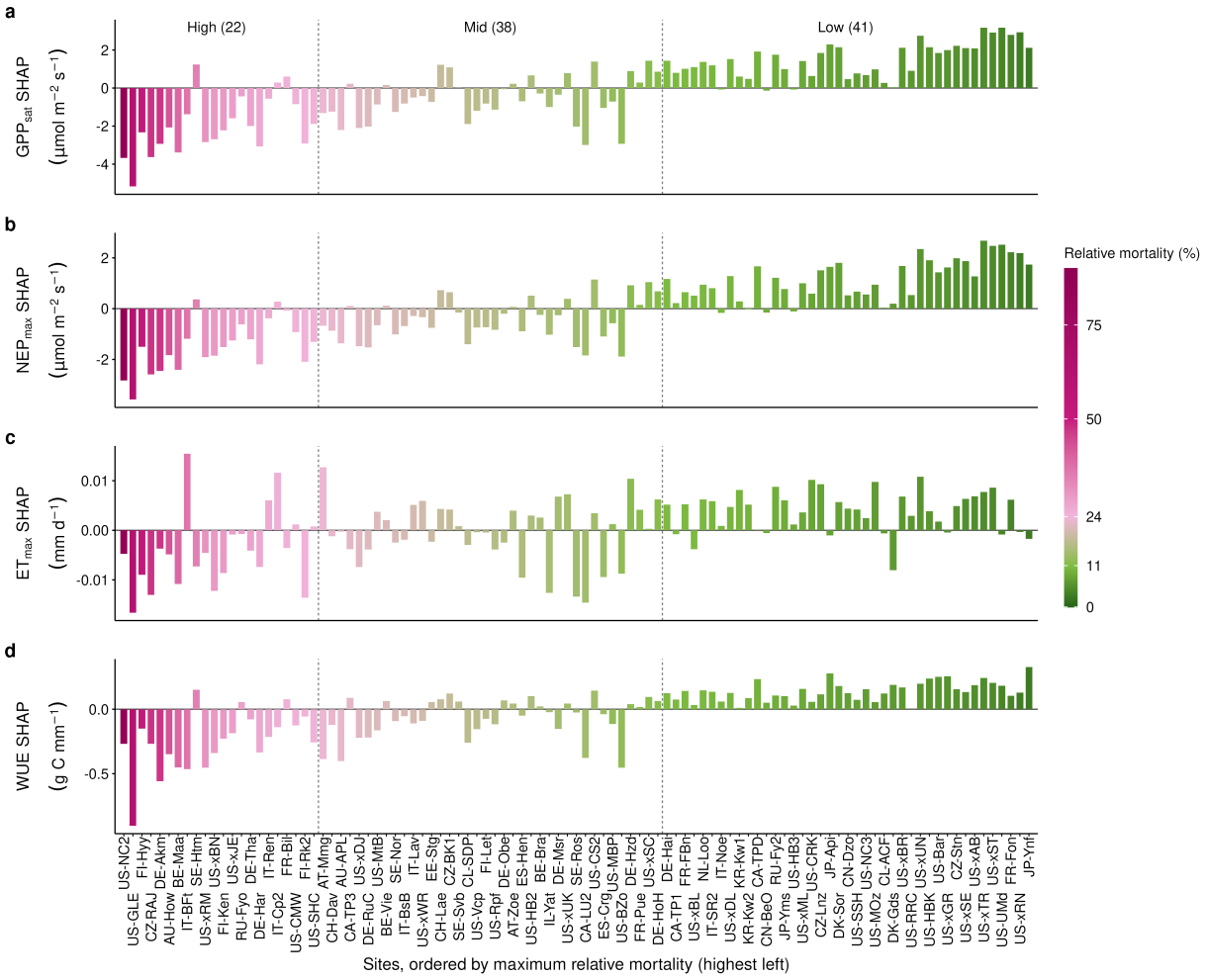



**Fig. 2** — net signed mortality SHAP per site (in the units of each EFP), C+T+M model, 24 months, AutoGluon; sites sorted by their maximum relative mortality (mean of the five buffers) and coloured from green (low) to pink (high), with the colour change anchored at the class breaks (10.6 and 24.0 %); FLUXNET site codes below the panels.

**Figure 3: Does the direction differ between High and Low+Mid sites?** (signed SHAP)

- **Below zero means the model predicts a LOWER value of the EFP** (not that it predicts less accurately). Classes: site maximum of relative mortality; High > mean + 0.5 SD (22 High vs 79 Low+Mid). Mann–Whitney U, BH-FDR across the figure.
- **A clear dose–response.** At **High**-mortality sites the block pushes predicted carbon uptake *down*; at **Low+Mid** sites slightly *up* (relative to the average site).
- $GPP_{sat}$: High −1.83 / −2.15, Low+Mid +0.47 / +0.66 $\mu$mol m$^{-2}$ s$^{-1}$ (12 / 24 m); $NEP_{max}$: High −1.61 / −1.51, Low+Mid +0.37 / +0.51; WUE: High −0.22 / −0.24, Low+Mid +0.07 g C mm$^{-1}$; $ET_{max}$: tiny (−0.003 / −0.005 mm d$^{-1}$). All 8 tests $q < 0.001$.
- **Read with Table 2:** mortality improves accuracy by lowering predictions at damaged sites. Sites are grouped by the same variables whose contribution is measured, so the test confirms

that the learned relationship is consistent; the direction and size are the informative part.

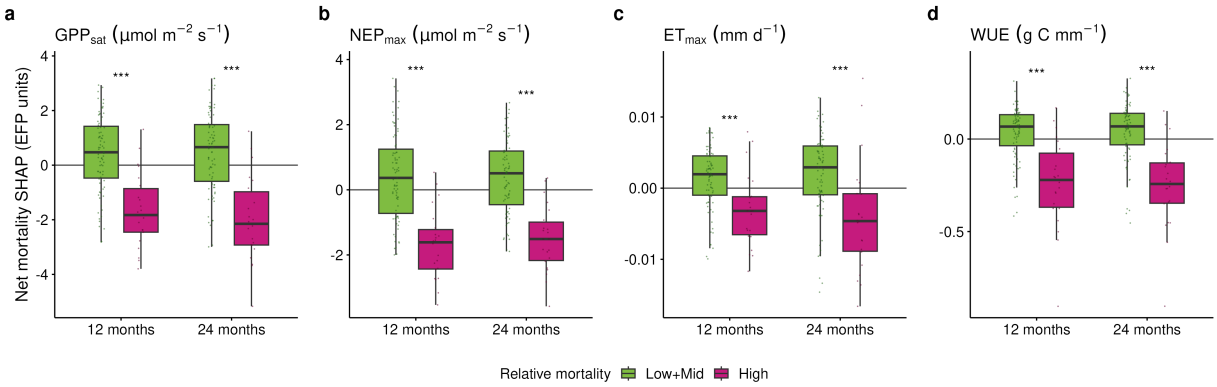



**Fig. 3** — net signed mortality SHAP, Low+Mid (green) vs High (pink) relative-mortality sites, AutoGluon, 12 and 24 months; colours as in Fig. 2.

**Figure 4: The direction tracks relative mortality**

- Per site, net signed mortality SHAP against the site's maximum relative mortality (AutoGluon, C+T+M, 24 months); linear fit with 95 % CI, Spearman test.
- **Strong and negative for the carbon fluxes and WUE:** $\rho = -0.81$ (GPP$_{sat}$), $-0.83$ (NEP$_{max}$), $-0.78$ (WUE); weaker for ET$_{max}$ ($-0.47$); all $p < 0.001$. More mortality, lower predicted carbon uptake.
- Sites with little mortality get a positive push, so the sign changes at roughly 10–20 % relative mortality.
- The benefit itself (error reduction) does not depend on tree cover ($\rho = 0.10$–$0.18$, n.s.), and the direction tracks mortality more closely than ecosystem type or tree cover (full board in the Appendix).

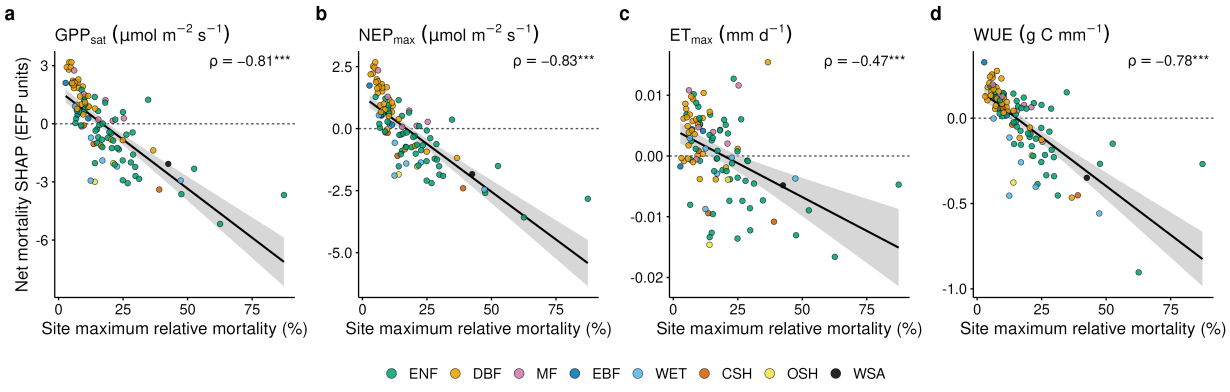



**Fig. 4** — relative mortality vs direction of the mortality effect, AutoGluon, C+T+M, 24 months; points coloured by IGBP class.

**Figure 5: Without mortality the model predicts too high in high-mortality years**

- The Low / Mid / High site groups lose power (22 High sites). Here every **site-year** is an observation, grouped by the relative mortality of that year; y = mean residual (observed − predicted), below 0 = the model predicts too high.
- **Without mortality (C+T)** the model over-predicts in the >30 % class (21 site-years, 10 sites): NEP$_{max}$ by $-4.0\,\mu$mol m$^{-2}$ s$^{-1}$ [−5.5, −2.4], GPP$_{sat}$ by −4.3 [−7.5, −1.3], WUE by

$-0.77\,\mathrm{g\,C\,mm^{-1}}$ [−1.10, −0.29] (AutoGluon, 24 months; 95 % site-block bootstrap). In the low classes the residual is close to 0.

- **With mortality (C+T+M)** the over-prediction shrinks by about two-thirds ($\mathrm{NEP}_{max}$ −1.3, $\mathrm{GPP}_{sat}$ −1.1, WUE −0.43) and the intervals include 0. $\mathrm{ET}_{max}$ shows no pattern.
- **The correction is significant.** Per site-year, correction = residual with − residual without mortality (= how much adding mortality lowers the prediction). In the >30 % class: $\mathrm{GPP}_{sat}$ +3.2, $\mathrm{NEP}_{max}$ +2.7, WUE +0.34 (all $q < 0.01$, paired permutation with sites swapped as blocks); in the 0–10 % class mortality slightly *raises* the predictions (−0.8, −0.7, −0.10). The correction grows with relative mortality (mixed model, per +10 %: $\mathrm{GPP}_{sat}$ +0.96, $\mathrm{NEP}_{max}$ +0.75, WUE +0.10; all $q < 0.001$, site-block bootstrap). XGBoost and LightGBM: the same, all significant, with smaller corrections (about half; Appendix A7).
- **Mixed model on the residuals** (residual ∼ relative mortality of the year, random intercept per site; all 361 site-years): without mortality the residual falls by −0.77 ($\mathrm{NEP}_{max}$), −0.18 (WUE) and −0.59 ($\mathrm{GPP}_{sat}$) per +10 % relative mortality; robust to a site-block bootstrap, to leaving out any single site and to removing the three sites with >50 % relative mortality. With mortality the trend is no longer significant (Appendix A7).
- The result comes mostly from the few site-years with >30 % relative mortality — the years in which mortality happened.

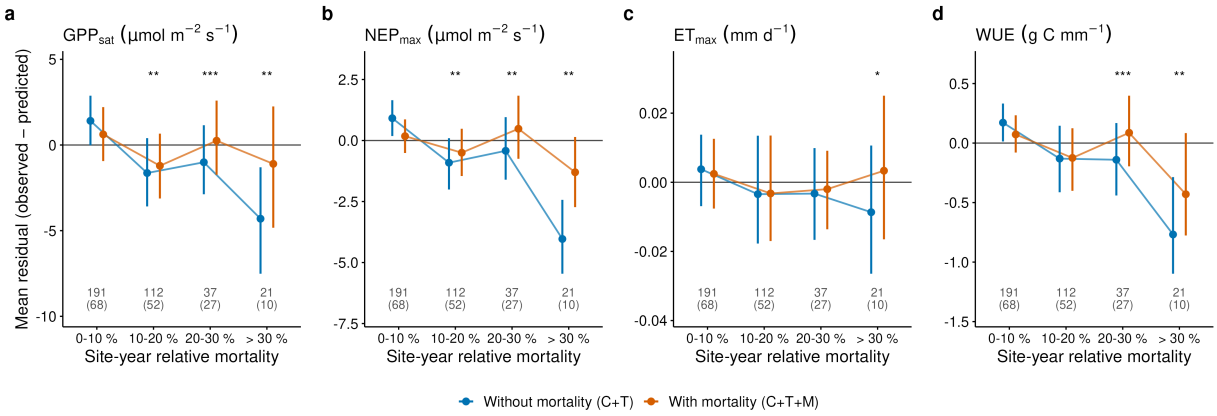



**Fig. 5** — mean residual per class of the site-year's relative mortality, without (blue) and with mortality (vermillion), AutoGluon, 24 months; bars = 95 % site-block bootstrap interval; numbers = site-years (sites); stars = adding mortality lowers the prediction significantly in that class (paired permutation, BH-FDR).

**Circularity check.** Refitting each property without the climate variables in its own formula (radiation for $\mathrm{GPP}_{sat}$, air temperature for $\mathrm{ET}_{max}$ and WUE) leaves the skill and the main mortality results essentially unchanged (Appendix A8).

### Mortality metrics

- The mortality block contains five metrics (absolute deadwood, relative deadwood, deadwood gain, relative tree-cover loss, relative mortality) at five buffers. They are strongly correlated and largely interchangeable: removing any single metric changes the error by at most about 1 % (drop-column test, Appendix).
- **Relative mortality carries most of the effect** (added alone to C+T it gives 6.7–7.1 % of the 7.6–8.9 % $\mathrm{NEP}_{max}$ error reduction of the full block) and has the largest SHAP share (26 %), so the figures use total mortality only. The other metrics are in the Appendix.

## Discussion

*Draft (about 1,000 words). Numbered citations; references in references.bib, each checked against the paper's abstract.*

Satellite- and aerial-based maps of tree mortality made the functional properties of woody ecosystems more predictable, and the gain was concentrated where mortality occurred. Without mortality information, models over-predicted maximum net ecosystem productivity, light-saturated photosynthesis and water-use efficiency in years with high mortality, and this bias grew with the amount of mortality. Adding mortality reduced it by one- to two-thirds, depending on the learner, and the direction of the mortality effect followed the amount of mortality rather than vegetation type or tree cover. Mortality therefore carries information on carbon uptake that climate and plant traits do not.

The direction of the effect is consistent with what single-site studies have measured during mortality events. At a subalpine forest hit by spruce beetles, net ecosystem exchange diminished by about half once the attacked trees died, after an earlier phase in which hydraulic failure restricted gas exchange[1]. Across North American forests, net ecosystem production became a carbon loss after stand-replacing disturbances and returned to a sink within 10 to 20 years[2]. Our results generalize these site observations across 101 sites and several ecosystem types, and they suggest why mortality acts on productivity: the first and most important axis of ecosystem function, maximum productivity, is mostly explained by vegetation structure[3], and mortality changes that structure directly. Relative mortality, which combines standing dead trees with tree-cover loss, carried most of the information, and the direction of its effect was negative at most sites with high mortality and slightly positive at sites with little.

The effect was not uniform, and the literature shows why. At a lodgepole pine forest where beetles killed 30 to 78 % of the basal area, maximum $CO_2$ uptake did not change, because the remaining vegetation used light more efficiently and the changes in photosynthesis and respiration cancelled out[4]. In British Columbia, beetle-killed stands remained growing-season carbon sinks thanks to surviving trees and understory[5], and a synthesis of remote-sensing and flux data suggested that net ecosystem production may recover within about four years after climate-driven mortality[6]. Such compensation fits the shape of our results: errors changed little at low and moderate mortality and fell sharply only in the site-years with more than 30 % relative mortality, and the reduction over all site-years (11–12 % for maximum net ecosystem productivity) was much smaller than in high-mortality years (29–32 %). Our predictors included mortality of the current and the one or two preceding years, which allows models to capture the multi-year legacies that follow drought and mortality[7]; how quickly the functional properties recover after mortality remains to be tested with longer records.

Mortality improved water-use efficiency where it occurred but not maximum evapotranspiration. Measured responses of evapotranspiration to tree death differ between sites: evapotranspiration fell by 28–36 % during a spruce beetle outbreak[1], and evapotranspiration relative to precipitation fell by 30 % at a beetle-affected site[8], whereas reviews of die-off ecohydrology expect the loss of transpiration and interception to be partly offset by more evaporation from the soil and by surviving and understory vegetation[9]. After drought-induced die-off in semi-arid basins, streamflow even decreased[10]. Maximum evapotranspiration, the 95th percentile of half-hourly evapotranspiration, is dominated by atmospheric demand and is therefore less sensitive to the loss of individual trees than carbon uptake. Mortality thus appears to act more on the carbon side of ecosystem function than on its water side.

These results matter for estimates of the forest carbon sink. Disturbances, many of them involving tree mortality, already shape the carbon balance at large scales: a mountain pine beetle outbreak turned the forests of British Columbia from a small carbon sink into a large source[11], increasing disturbances could reduce the carbon storage of European forests by about 500 Tg C in 2021–2030[12],

drought and fire brought the Siberian forest carbon sink close to neutral[13], and stand-replacing disturbances account for a large part of global biomass turnover[14]. Data-driven upscaling of eddy-covariance fluxes does not include time since disturbance or site history among its predictors, which likely contributes to biased net ecosystem exchange patterns[15], and the representation of disturbance in vegetation models is a key reason why their estimates of the northern carbon sink disagree with atmospheric inversions[16]. Our models reproduce this blind spot: without mortality they over-estimated carbon uptake exactly where mortality was high. With tree mortality increasing worldwide[17–19] and forest disturbance regimes shifting towards younger and shorter stands[20,21], this bias is likely to grow unless mortality is represented explicitly.

Our study has limitations. The models are predictive, so they quantify how much mortality information improves predictions rather than the physiological mechanisms behind the change. Only 21 site-years at 10 sites had more than 30 % relative mortality; the strongest effects rest on these sites, although the over-prediction without mortality held when any single site or the three most extreme sites were left out. Mortality was derived from maps around each tower (100–500 m buffers), which do not match the varying flux footprint exactly, and errors in the maps add noise that probably weakens rather than strengthens the effects we report. Finally, the climate predictors cover only the period before the growing season, and leaving out the climate variables that enter the calculation of each property did not change the main results (Appendix A8), so the mortality effect does not arise from circularity with the climate data.

Knowing where and when trees die makes the carbon uptake of woody ecosystems more predictable. Without that knowledge, models systematically over-estimated carbon uptake in ecosystems with high mortality; with it, this bias shrank, and the correction scaled with the amount of mortality regardless of vegetation type or tree cover. Mortality maps now cover large areas at high resolution[22], and flux networks keep growing. Adding mortality to data-driven upscaling products and to the evaluation of land-surface models is a direct way to avoid over-estimating the carbon uptake of woody ecosystems with high mortality.

## Methods

**Eddy-covariance data.** We used half-hourly FLUXNET data products processed with ONEFlux v1.3[23], downloaded for all available sites with the FLUXNET Shuttle on 12 May 2026. Net ecosystem exchange (NEE_VUT_REF) and gross primary productivity from night-time partitioning (GPP_NT_VUT_REF[24]) were used together with the tower meteorology (air temperature, vapour pressure deficit, incoming shortwave radiation, precipitation) and latent heat flux. Only measured or high-quality gap-filled half-hours (quality flags 0 and 1) entered the calculation of the ecosystem functional properties. The 101 sites of the final dataset come from AmeriFlux (44 sites), ICOS (29), the European Fluxes Database (18), JapanFlux[25] (3), KoFlux (2), ChinaFlux[26] (2), TERN-OzFlux[27,28] (2) and FLUXNET (1); the data of each site are cited individually (Data availability).

**Ecosystem functional properties.** For each site-year we calculated four ecosystem functional properties following Migliavacca et al.[3], within the growing season, defined as the days on which daily gross primary productivity exceeded 30 % of its 95th percentile (bigleaf R package[29]). Light-saturated gross primary productivity ($GPP_{sat}$) was obtained by fitting a light-response curve of NEE against photosynthetic photon flux density (shortwave radiation × 2.11) in five-day windows of daytime data (shortwave radiation > 50 $W\,m^{-2}$) and taking the 95th percentile of the fitted saturation values per year[3,30]. Maximum net ecosystem productivity ($NEP_{max}$) is the 90th percentile of all half-hourly net ecosystem productivity (−NEE, day and night) in the growing season[3]. Maximum evapotranspiration ($ET_{max}$) is the 95th percentile of half-hourly evapotranspiration ($mm\,d^{-1}$), and water-use efficiency (WUE) the median ratio of gross primary productivity to evapotranspiration ($g\,C\,mm^{-1}$)[29]; both used daytime half-hours with shortwave radiation > 200 $W\,m^{-2}$ and friction velocity > 0.2 $m\,s^{-1}$ and excluded half-hours within 24 h after precipitation of more than 0.1 mm, so

that evaporation of intercepted water does not inflate evapotranspiration. Evapotranspiration was converted from latent heat flux with the air-temperature-dependent latent heat of vaporization[29].

**Tree mortality.** Tree mortality was derived from annual maps of standing deadwood cover and tree cover produced by deadtrees.earth from Sentinel-2 imagery, trained on centimetre-resolution aerial imagery with manually annotated dead trees[22] (map version 5; year 2016 excluded as unreliable). Within five nested circular buffers around each tower (radius 100, 200, 300, 400 and 500 m; pixels without a valid prediction masked) we calculated, per year: absolute deadwood (deadwood cover as a percentage of buffer area), relative deadwood (deadwood cover relative to tree cover, where tree cover $\geq$ 10 %, capped at 100 %), deadwood gain (year-on-year increase of deadwood cover, percentage points), relative tree-cover loss (percentage of the previous year's tree cover lost) and relative mortality, (deadwood + tree-cover loss) / (tree cover + tree-cover loss) $\times$ 100. Each metric entered the models for every buffer, for the year of the ecosystem functional property and for the one (12-month window) or two (24-month window) preceding years. For figures, sites were classified by their maximum relative mortality across years, averaged over the five buffers (Low $<$ mean $-$ 0.5 s.d., High $>$ mean + 0.5 s.d. across sites).

**Climate and plant traits.** Climate predictors were calculated from the tower meteorology and cover only the period before the growing season: the 12 months ending in the month before the growing-season start (lag 1) and, for the 24-month window, the 12 months before that (lag 2). For every month we used the mean, 5th and 95th percentile of air temperature, vapour pressure deficit and shortwave radiation, and the precipitation sum. Plant traits were 23 community-weighted mean traits at 1 km resolution from global maps based on crowdsourced biodiversity observations and Earth observation[31], averaged within 500 m of each tower and weighted by area.

**Site selection.** Site-years were retained when all four ecosystem functional properties were available (2017–2025), lay in runs of at least two consecutive years and had complete climate and mortality predictors including their lags, which restricts the years to 2020–2025. Two plantation sites with extreme water-use efficiency (CN-Sb1, CN-Sb2) and site-years with negative water-use efficiency were removed. We kept sites with a mean tree cover of at least 30 % within 500 m. The final dataset has 101 sites and 361 site-years (median 4 years per site) of evergreen needleleaf (49 sites), deciduous broadleaf (30), mixed (7) and evergreen broadleaf (3) forests, wetlands (6), closed (3) and open shrublands (2) and woody savanna (1).

**Models.** For each property and window, two models were compared that differ only in the mortality predictors: climate and traits (C+T; 143 predictors for 12 months, 263 for 24 months) and climate, traits and mortality (C+T+M; 193 and 338 predictors). The main learner was AutoGluon-Tabular[32] (v1.6.3; default model zoo of gradient-boosted trees, random forests, extra trees and neural networks, five-fold bagging on site-disjoint folds, weighted ensemble without stacking). XGBoost[33] and LightGBM[34], each tuned with Optuna[35] (tree-structured Parzen estimator, 50 trials per property and window, five-fold cross-validation on site-disjoint groups), served as checks.

**Cross-validation.** Models were evaluated with repeated leave-one-site-out cross-validation: for every site, models were trained three times on a random 80 % of the remaining sites and the predictions for the held-out site were averaged. AutoGluon was refitted inside every fold, so that model selection never saw the held-out site.

**Model interpretation.** Contributions of predictors were quantified with SHAP values[36]: exact TreeSHAP[37] for XGBoost and LightGBM and permutation-based SHAP for the AutoGluon ensemble. The net contribution of the mortality block is the sum of the signed SHAP values of all mortality predictors per site-year, averaged per site; negative values mean that mortality lowers the predicted property.

**Statistics.** The main test compares the mean absolute error (MAE) of C+T+M and C+T at the site-year level: every site-year is an observation, and the statistic is the difference of the pooled MAE (percentage change = ratio of the pooled errors). Because the years of one site are not

independent, whole sites were resampled with replacement (10 000 bootstrap draws, 95 % percentile intervals) and, for a paired permutation test, the C+T and C+T+M predictions were swapped per site (10 000 draws, one-sided). The same tests were run for subsets of sites (High relative mortality) and of site-years (relative mortality of the year above the Mid or High break). At the site level, per-site errors were compared with a one-sided sign-flip permutation test on the log error ratios (100 000 flips). The relationship between the residuals (observed − predicted) and the total mortality of the site-year was tested with linear mixed models with a random intercept per site (nlme[38]), checked with a site-block bootstrap of the slope (1,000 draws), leave-one-site-out refits and without the three sites with more than 50 % relative mortality; the correction of the residual by mortality (residual of C+T+M minus residual of C+T) was tested per mortality class with the paired site-level permutation (100 000 draws). Differences in net mortality SHAP between High and Low+Mid sites were tested with Mann–Whitney U tests and associations with Spearman correlations. $P$ values were corrected for multiple testing with the Benjamini–Hochberg procedure[39] within each table or figure. To test for circularity, each property was refitted without the climate variables that enter its own formula (shortwave radiation for $GPP_{sat}$, air temperature for $ET_{max}$ and WUE).

**Software.** Analyses used R[40] (v4.3.3; xgboost, lightgbm, nlme, bigleaf) and Python (AutoGluon, Optuna, shap).

**Use of large language models.** **[DECISION: wording to be agreed by the authors; the journal requires any use of LLMs (e.g. for code or text assistance) to be documented here]**

## Data availability

The FLUXNET data products used in this work were downloaded from the ICOS Carbon Portal and AmeriFlux via the FLUXNET Shuttle on 12 May 2026 and are cited per site[41–141]. Tree mortality maps are available from deadtrees.earth[22] and plant trait maps from Zenodo[31]. The derived site-year dataset and all model predictions are deposited at [repository, DOI].

## Code availability

All code for dataset construction, model training (AutoGluon, XGBoost, LightGBM), statistics and figures is available at [repository, DOI].

*Discussion references so far; Main and Methods still to be added (about 50 for an Article).*

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# Extended Data and Supplementary Information

*The journal allows at most 10 Extended Data items (each cited in the main text); everything else goes to Supplementary Information. The A-sections below still need to be split into Extended Data and Supplementary.*

## A1. Learner comparison

- Same data, predictors, windows and cross-validation for all three learners.
- AutoGluon best in 7 of 16 tasks; mean $R^2$ (C+T+M) 0.402 / 0.408 (12 / 24 m) vs XGBoost 0.385 / 0.406 and LightGBM 0.388 / 0.400.
- The effect of mortality is very similar: $\mathrm{NEP}_{max}$ improves for every learner; overall RMSE change for $\mathrm{GPP}_{sat}$ −2.8 to −4.0 % for all learners; AutoGluon gives a somewhat larger WUE effect (−4 % vs −1 to −4 %).

Table 3: Skill of the C+T+M model under repeated leave-one-site-out CV ($R^2$), all learners. Bottom rows: mean over the four EFPs.

| | 12 months | | | 24 months | | |
|---|---|---|---|---|---|---|
| EFP | AutoGluon | XGBoost | LightGBM | AutoGluon | XGBoost | LightGBM |
| $\mathrm{GPP_{sat}}$ | **0.367** | 0.334 | 0.346 | 0.370 | **0.387** | 0.347 |
| $\mathrm{NEP_{max}}$ | **0.503** | 0.473 | 0.492 | 0.502 | 0.514 | **0.540** |
| $\mathrm{ET_{max}}$ | 0.425 | 0.439 | **0.443** | 0.433 | 0.447 | **0.453** |
| WUE | **0.312** | 0.292 | 0.270 | **0.326** | 0.277 | 0.261 |
| Mean, C+T+M | **0.402** | 0.385 | 0.388 | **0.408** | 0.406 | 0.400 |
| Mean, C+T | **0.349** | 0.337 | 0.347 | 0.351 | **0.365** | 0.356 |

## A2. Per-site error: figures and site-level tests

- Per-site MAE without and with mortality; % = geometric-mean change of the per-site MAE, stars = one-sided paired sign-flip permutation test on the per-site log MAE ratios (100 000 flips, BH-FDR across the figure). The log ratio stops sites with an almost perfect C+T prediction from dominating (one-year sites, e.g. KR-Kw1).
- $\mathrm{NEP}_{max}$ MAE −12 / −8 % across all sites and −30 / −17 % at High sites (12 / 24 months); WUE −11 / −15 % at High sites.




**Fig. A1** — per-site MAE without (cyan) and with mortality (pink), AutoGluon, all 101 sites.

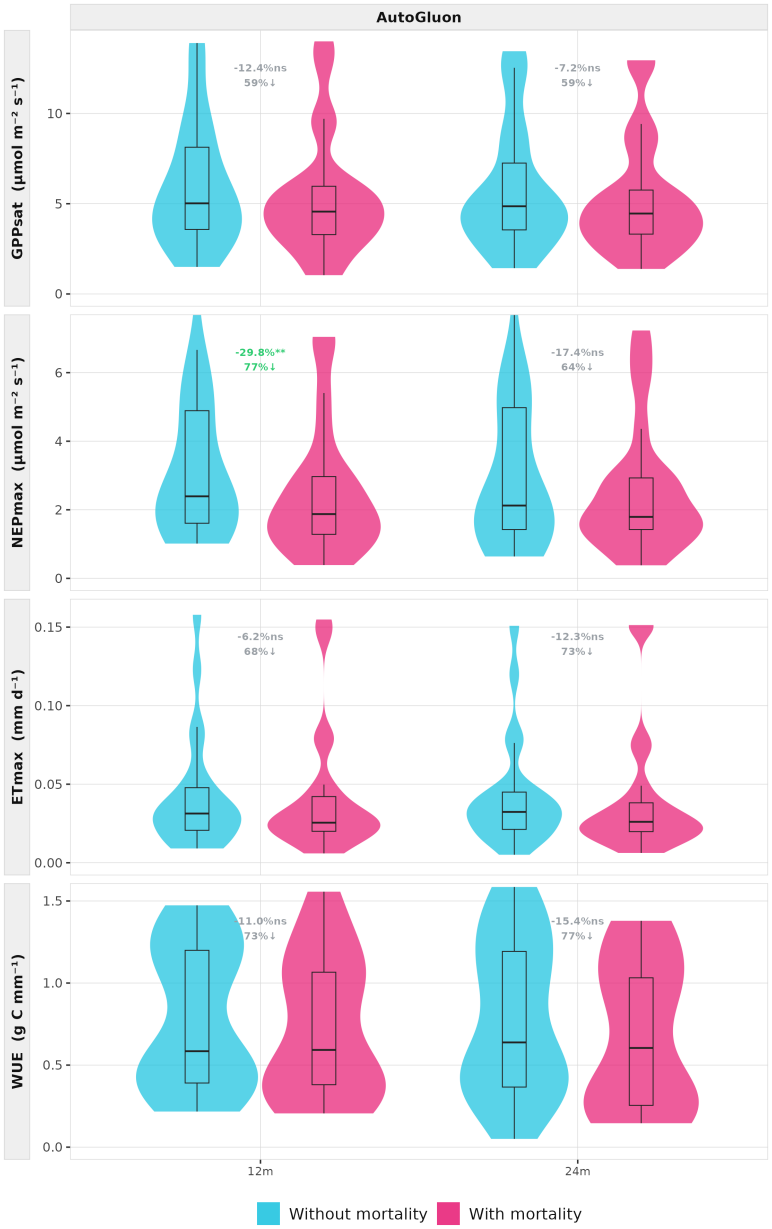



**Fig. A2** — the same, high relative-mortality sites only (22 sites; models trained on all sites).

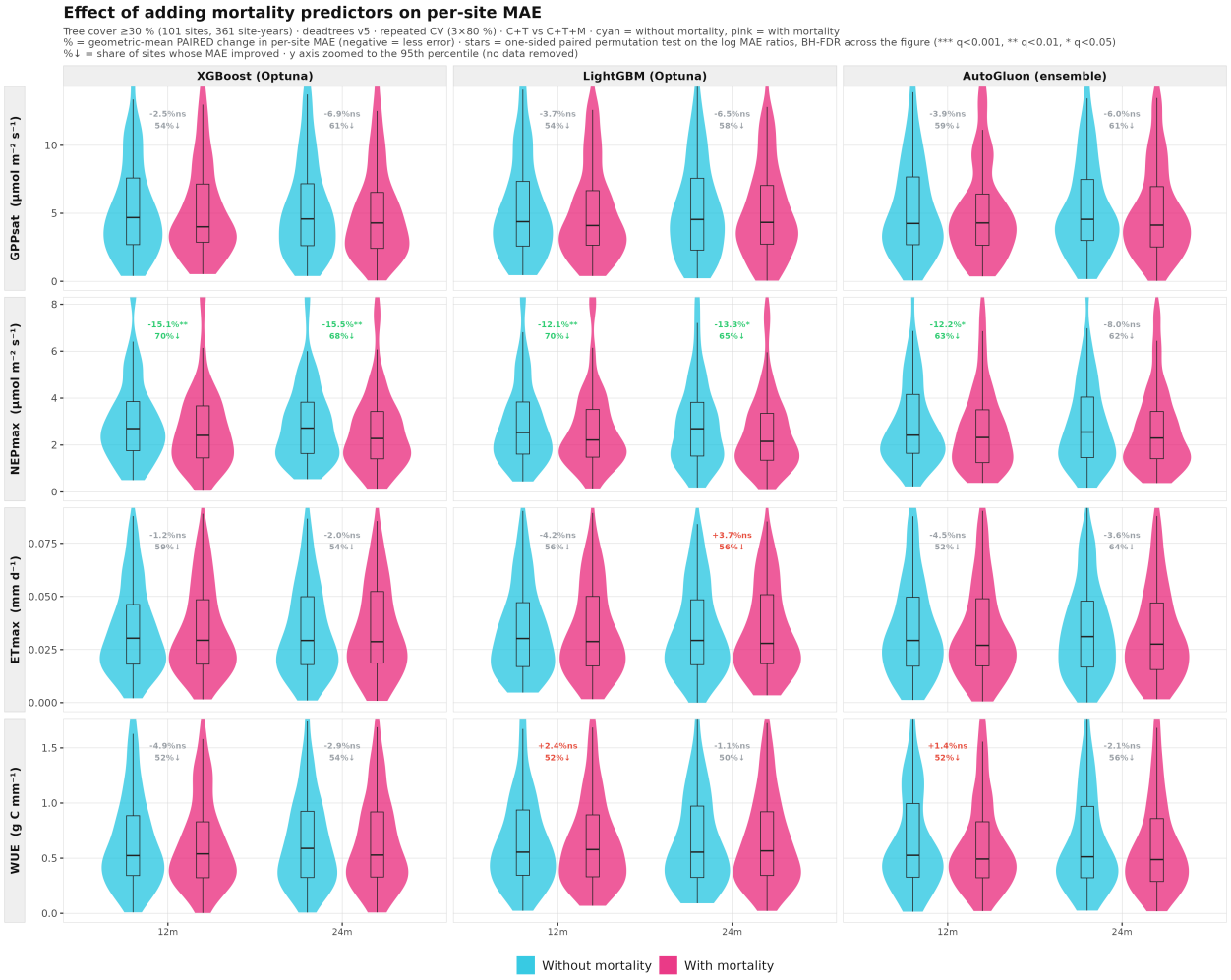



**Fig. A3** — Fig. A1 for all three learners.

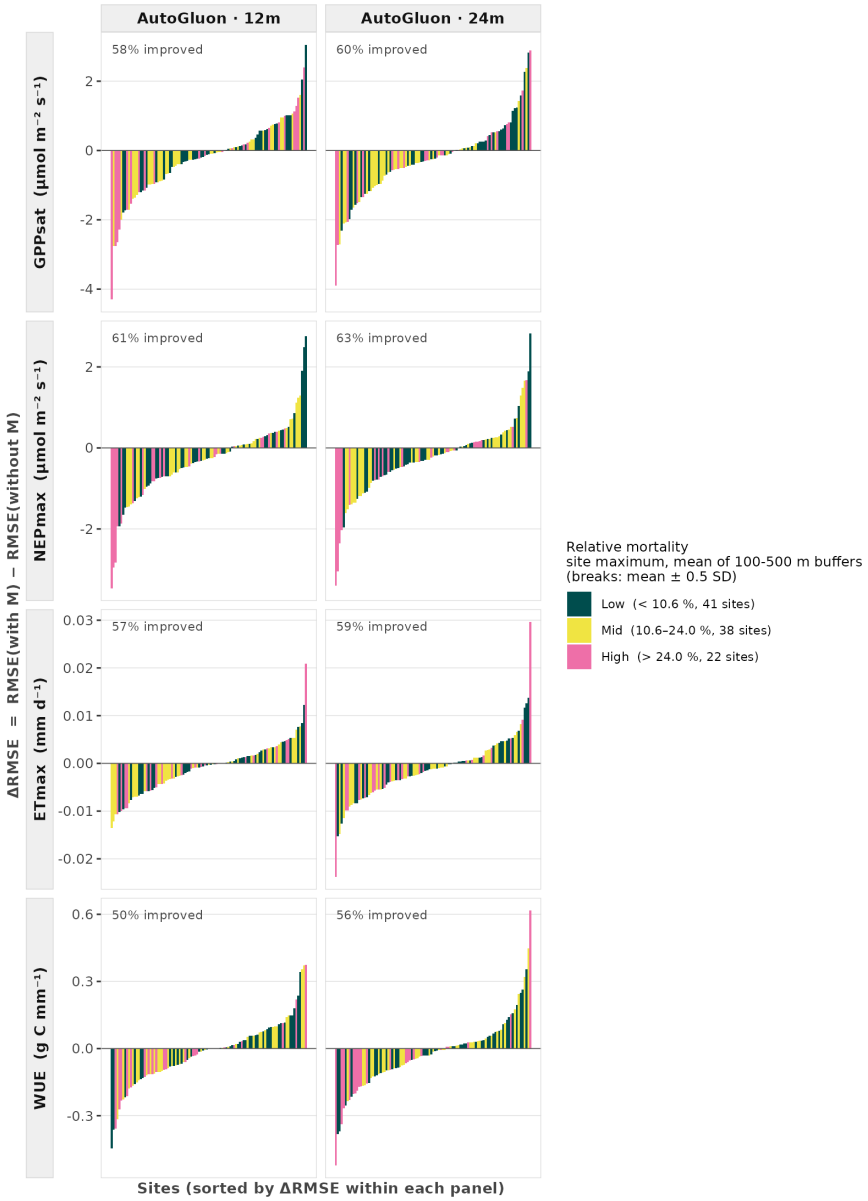



**Fig. A4** — per-site ΔRMSE, AutoGluon, bars coloured by relative-mortality class.

Table 4: Effect of adding mortality predictors on per-site prediction error (summary of Fig. 2). For every site, MAE of the model with climate, traits and mortality (C+T+M) is compared with the model without mortality (C+T), both evaluated by repeated leave-one-site-out cross-validation ($3 \times 80\%$). $\Delta$MAE = geometric mean over sites of the paired error ratio, $(\exp(\text{mean}_{\text{sites}} \ln(\text{MAE}_{\text{C+T+M}} / \text{MAE}_{\text{C+T}})) - 1) \times 100$; negative values mean that mortality reduces the error. Improved = share of sites whose MAE decreased. Sites are grouped by their maximum relative mortality across years (averaged over the five buffers, 100–500 m); High > mean + 0.5 SD (24.0%), Low < mean − 0.5 SD (10.6%), Mid in between. Stars: one-sided paired permutation test on the per-site log error ratios (100 000 random sign flips; C+T+M < C+T), Benjamini–Hochberg corrected across all 72 tests of the three learners ($^{*}\, q < 0.05$, $^{**}\, q < 0.01$, $^{***}\, q < 0.001$). Tree cover ≥ 30%, 101 sites, 361 site-years, deadtrees v5.

| Model | EFP | | All sites ($n = 101$) | Mid + High ($n = 60$) | High ($n = 22$) |
|---|---|---|---|---|---|
| AutoGluon 12 months | $GPP_{sat}$ | $\Delta$MAE (%) | $-3.9$ | $-8.6^{*}$ | $-12.4$ |
| | | Improved (%) | 59 | 62 | 59 |
| | $NEP_{max}$ | $\Delta$MAE (%) | $-12.2^{*}$ | $-14.5^{**}$ | $-29.8^{**}$ |
| | | Improved (%) | 63 | 62 | 77 |
| | $ET_{max}$ | $\Delta$MAE (%) | $-4.5$ | $-5.6$ | $-6.2$ |
| | | Improved (%) | 52 | 58 | 68 |
| | WUE | $\Delta$MAE (%) | $+1.4$ | $-5.2$ | $-11.0$ |
| | | Improved (%) | 52 | 60 | 73 |
| AutoGluon 24 months | $GPP_{sat}$ | $\Delta$MAE (%) | $-6.0$ | $-9.0^{**}$ | $-7.2$ |
| | | Improved (%) | 61 | 72 | 59 |
| | $NEP_{max}$ | $\Delta$MAE (%) | $-8.0$ | $-9.1$ | $-17.4$ |
| | | Improved (%) | 62 | 57 | 64 |
| | $ET_{max}$ | $\Delta$MAE (%) | $-3.6$ | $-6.9^{*}$ | $-12.3$ |
| | | Improved (%) | 64 | 72 | 73 |
| | WUE | $\Delta$MAE (%) | $-2.1$ | $-6.1$ | $-15.4$ |
| | | Improved (%) | 56 | 63 | 77 |

Table 5: Paired bootstrap of the change in MAE when mortality is added (C+T → C+T+M), AutoGluon. Cells: % change of the error, $100\,(\text{MAE}_{\text{C+T+M}} / \text{MAE}_{\text{C+T}} - 1)$, with the 95 % bootstrap interval in brackets. *Site level*: one MAE per site and model over its years, ratio of the mean site errors. *Site-year level*: every site-year is an observation (pooled MAE; no averaging over years). Years of one site are not independent, so whole sites are resampled (10 000 bootstrap draws) and the C+T / C+T+M labels are swapped per site for the paired permutation test (10 000 draws, one-sided; stars, Benjamini–Hochberg corrected across all three learners: $^{*}\, q < 0.05$, $^{**}\, q < 0.01$, $^{***}\, q < 0.001$). High sites: site maximum of relative mortality > 24.0 %; the last two columns keep only the site-years whose own relative mortality exceeds 10.6 % (Mid + High) or 24.0 % (High). Column heads: sites / site-years. 101 sites, 361 site-years, deadtrees v5.

| | | Site level | Site-year level | | | |
|---|---|---|---|---|---|---|
| | | All sites (101 sites) | All sites (101 / 361) | High sites (22 / 80) | Mid + High years (60 / 160) | High years (22 / 40) |
| AutoGluon 12 months | $GPP_{sat}$ | $-4.6$ [$-8.6$, $-0.5$]$^{*}$ | $-4.1$ [$-8.0$, $-0.2$] | $-7.2$ [$-18.1$, $+2.9$] | $-6.7$ [$-12.8$, $-0.7$]$^{*}$ | $-12.0$ [$-29.5$, $+4.9$] |
| | $NEP_{max}$ | $-10.1$ [$-15.5$, $-4.4$]$^{**}$ | $-11.8$ [$-17.0$, $-6.3$]$^{**}$ | $-22.6$ [$-33.6$, $-11.1$]$^{**}$ | $-12.5$ [$-20.1$, $-4.9$]$^{*}$ | $-31.6$ [$-43.2$, $-16.0$]$^{*}$ |
| | $ET_{max}$ | $-2.5$ [$-5.6$, $+0.5$] | $-1.3$ [$-4.2$, $+1.3$] | $-2.9$ [$-8.4$, $+2.1$] | $-3.9$ [$-7.7$, $+0.0$] | $-9.7$ [$-20.6$, $+2.7$] |
| | WUE | $-3.2$ [$-7.3$, $+1.3$] | $-4.3$ [$-8.1$, $-0.2$] | $-10.9$ [$-17.8$, $-4.3$]$^{*}$ | $-7.2$ [$-12.2$, $-1.8$]$^{*}$ | $-16.9$ [$-25.2$, $-5.1$]$^{*}$ |
| AutoGluon 24 months | $GPP_{sat}$ | $-5.5$ [$-9.4$, $-1.4$]$^{*}$ | $-5.7$ [$-9.5$, $-1.7$]$^{*}$ | $-7.1$ [$-17.2$, $+3.6$] | $-8.3$ [$-14.2$, $-2.6$]$^{*}$ | $-11.1$ [$-28.0$, $+8.1$] |
| | $NEP_{max}$ | $-9.3$ [$-14.4$, $-3.7$]$^{**}$ | $-10.8$ [$-16.0$, $-5.1$]$^{**}$ | $-17.5$ [$-30.7$, $-2.1$] | $-11.6$ [$-19.8$, $-2.7$]$^{*}$ | $-29.2$ [$-42.9$, $-8.6$]$^{*}$ |
| | $ET_{max}$ | $-3.8$ [$-7.3$, $-0.1$]$^{*}$ | $-2.0$ [$-5.0$, $+0.9$] | $-5.3$ [$-13.1$, $+1.2$] | $-5.4$ [$-9.6$, $-1.3$]$^{*}$ | $-11.4$ [$-23.7$, $+3.2$] |
| | WUE | $-3.6$ [$-8.3$, $+1.6$] | $-4.7$ [$-9.1$, $-0.1$] | $-14.8$ [$-23.9$, $-6.4$]$^{*}$ | $-8.2$ [$-14.5$, $-1.5$]$^{*}$ | $-21.8$ [$-34.5$, $-8.2$]$^{*}$ |

Table 6: Paired bootstrap of the change in RMSE when mortality is added (C+T → C+T+M), AutoGluon. Cells: % change of the error, $100\,(\mathrm{RMSE}_{\mathrm{C+T+M}} / \mathrm{RMSE}_{\mathrm{C+T}} - 1)$, with the 95 % bootstrap interval in brackets. *Site level*: one RMSE per site and model over its years, ratio of the mean site errors. *Site-year level*: every site-year is an observation (pooled RMSE; no averaging over years). Years of one site are not independent, so whole sites are resampled (10 000 bootstrap draws) and the C+T / C+T+M labels are swapped per site for the paired permutation test (10 000 draws, one-sided; stars, Benjamini–Hochberg corrected across all three learners: $^{*}\,q < 0.05$, $^{**}\,q < 0.01$, $^{***}\,q < 0.001$). High sites: site maximum of relative mortality > 24.0 %; the last two columns keep only the site-years whose own relative mortality exceeds 10.6 % (Mid + High) or 24.0 % (High). Column heads: sites / site-years. 101 sites, 361 site-years, deadtrees v5.

| | | Site level | Site-year level | | | |
|---|---|---|---|---|---|---|
| | | All sites (101 sites) | All sites (101 / 361) | High sites (22 / 80) | Mid + High years (60 / 160) | High years (22 / 40) |
| AutoGluon 12 months | $\mathrm{GPP}_{\mathrm{sat}}$ | −3.6 [−7.2, +0.1] | −2.8 [−6.1, +0.5] | −4.4 [−15.2, +4.5] | −4.4 [−10.0, +0.9] | −10.1 [−27.6, +5.8] |
| | $\mathrm{NEP}_{\mathrm{max}}$ | −9.3 [−14.4, −3.9]$^{**}$ | −9.7 [−14.6, −4.7]$^{**}$ | −21.0 [−33.7, −7.7]$^{*}$ | −11.8 [−19.7, −4.3]$^{*}$ | −32.6 [−45.1, −16.0]$^{*}$ |
| | $\mathrm{ET}_{\mathrm{max}}$ | −2.4 [−5.1, +0.3] | −0.2 [−2.6, +2.1] | +0.0 [−5.9, +5.6] | −0.7 [−4.7, +3.5] | −0.7 [−17.2, +17.2] |
| | WUE | −2.2 [−6.0, +1.8] | −4.1 [−7.3, −1.2]$^{*}$ | −7.5 [−14.3, −1.5]$^{*}$ | −5.7 [−10.3, −1.4]$^{*}$ | −14.3 [−22.3, −2.7] |
| AutoGluon 24 months | $\mathrm{GPP}_{\mathrm{sat}}$ | −4.3 [−7.9, −0.6]$^{*}$ | −4.0 [−7.3, −0.2]$^{*}$ | −4.2 [−14.0, +4.7] | −6.0 [−11.2, −1.2]$^{*}$ | −9.8 [−25.4, +7.0] |
| | $\mathrm{NEP}_{\mathrm{max}}$ | −8.3 [−13.3, −2.9]$^{**}$ | −9.4 [−14.2, −4.2]$^{*}$ | −19.5 [−32.3, −5.7]$^{*}$ | −11.8 [−19.5, −3.8]$^{*}$ | −31.0 [−44.0, −13.1]$^{*}$ |
| | $\mathrm{ET}_{\mathrm{max}}$ | −3.1 [−6.3, +0.2] | −0.4 [−3.1, +2.6] | +0.6 [−7.8, +7.9] | −1.6 [−6.4, +2.9] | +0.4 [−19.3, +22.0] |
| | WUE | −3.0 [−7.2, +1.6] | −4.3 [−7.4, −1.2]$^{*}$ | −9.2 [−16.3, −2.4]$^{*}$ | −6.7 [−11.4, −1.5]$^{*}$ | −16.9 [−24.6, −6.5]$^{*}$ |

Table 7: Paired bootstrap of the change in MAE when mortality is added (C+T → C+T+M), XGBoost and LightGBM. Cells: % change of the error, $100\,(\mathrm{MAE}_{\mathrm{C+T+M}} / \mathrm{MAE}_{\mathrm{C+T}} - 1)$, with the 95 % bootstrap interval in brackets. *Site level*: one MAE per site and model over its years, ratio of the mean site errors. *Site-year level*: every site-year is an observation (pooled MAE; no averaging over years). Years of one site are not independent, so whole sites are resampled (10 000 bootstrap draws) and the C+T / C+T+M labels are swapped per site for the paired permutation test (10 000 draws, one-sided; stars, Benjamini–Hochberg corrected across all three learners: $^{*}\,q < 0.05$, $^{**}\,q < 0.01$, $^{***}\,q < 0.001$). High sites: site maximum of relative mortality > 24.0 %; the last two columns keep only the site-years whose own relative mortality exceeds 10.6 % (Mid + High) or 24.0 % (High). Column heads: sites / site-years. 101 sites, 361 site-years, deadtrees v5.

| | | Site level | Site-year level | | | |
|---|---|---|---|---|---|---|
| | | All sites (101 sites) | All sites (101 / 361) | High sites (22 / 80) | Mid + High years (60 / 160) | High years (22 / 40) |
| XGBoost 12 months | $\mathrm{GPP}_{\mathrm{sat}}$ | −3.5 [−6.6, −0.3]$^{*}$ | −3.2 [−6.3, +0.1] | −5.3 [−12.7, +2.5] | −3.7 [−8.1, +0.9] | −10.6 [−18.1, +0.7] |
| | $\mathrm{NEP}_{\mathrm{max}}$ | −10.3 [−15.1, −5.0]$^{**}$ | −10.7 [−15.7, −4.9]$^{**}$ | −17.4 [−27.7, −7.7]$^{**}$ | −12.3 [−18.8, −5.8]$^{**}$ | −28.8 [−37.3, −17.6]$^{**}$ |
| | $\mathrm{ET}_{\mathrm{max}}$ | −1.4 [−3.5, +0.7] | −0.9 [−3.0, +1.3] | +0.1 [−2.9, +3.8] | −2.4 [−5.0, +0.2] | −1.9 [−6.1, +2.7] |
| | WUE | −2.8 [−6.9, +1.7] | −2.9 [−6.3, +0.7] | −6.9 [−13.6, +0.5] | −4.2 [−8.8, +0.9] | −12.7 [−19.2, −5.9]$^{*}$ |
| XGBoost 24 months | $\mathrm{GPP}_{\mathrm{sat}}$ | −5.8 [−9.9, −1.6]$^{*}$ | −5.4 [−9.8, −1.0]$^{*}$ | −6.6 [−16.2, +2.9] | −7.8 [−13.4, −2.4]$^{*}$ | −6.4 [−18.2, +5.1] |
| | $\mathrm{NEP}_{\mathrm{max}}$ | −10.1 [−14.6, −5.6]$^{**}$ | −9.5 [−14.0, −4.8]$^{**}$ | −17.2 [−27.9, −8.3]$^{**}$ | −9.7 [−15.5, −4.2]$^{**}$ | −19.8 [−30.6, −10.7]$^{**}$ |
| | $\mathrm{ET}_{\mathrm{max}}$ | −0.3 [−2.8, +2.2] | −0.4 [−3.0, +2.1] | −2.0 [−5.9, +1.7] | −2.5 [−5.6, +0.7] | −4.0 [−8.2, +0.6] |
| | WUE | −1.8 [−6.3, +3.3] | −2.3 [−6.1, +2.1] | −5.3 [−12.8, +4.6] | −3.1 [−8.3, +3.5] | −11.7 [−19.7, −3.1]$^{*}$ |
| LightGBM 12 months | $\mathrm{GPP}_{\mathrm{sat}}$ | −4.0 [−7.1, −0.8]$^{*}$ | −3.3 [−6.4, +0.1] | −6.6 [−14.0, +0.4] | −5.8 [−10.0, −1.6]$^{*}$ | −10.3 [−17.7, −3.1]$^{*}$ |
| | $\mathrm{NEP}_{\mathrm{max}}$ | −9.5 [−14.8, −3.8]$^{**}$ | −9.6 [−15.0, −3.4]$^{**}$ | −14.7 [−25.4, −4.1]$^{*}$ | −10.7 [−17.5, −3.1]$^{*}$ | −24.7 [−35.2, −11.3]$^{*}$ |
| | $\mathrm{ET}_{\mathrm{max}}$ | −2.1 [−4.4, +0.3] | −1.4 [−3.7, +1.0] | −0.2 [−4.0, +4.6] | −3.4 [−6.4, −0.4]$^{*}$ | −3.8 [−9.5, +2.4] |
| | WUE | +0.0 [−3.7, +4.2] | −1.6 [−4.7, +2.2] | −2.8 [−10.0, +6.5] | −2.8 [−7.6, +3.1] | −7.9 [−15.0, +0.4] |
| LightGBM 24 months | $\mathrm{GPP}_{\mathrm{sat}}$ | −3.8 [−6.7, −0.8]$^{*}$ | −3.1 [−6.3, +0.1] | −5.2 [−12.0, +1.8] | −4.3 [−8.6, +0.1] | −5.6 [−15.4, +5.9] |
| | $\mathrm{NEP}_{\mathrm{max}}$ | −12.2 [−17.6, −6.6]$^{**}$ | −11.4 [−16.6, −5.7]$^{**}$ | −14.0 [−24.9, −4.0]$^{*}$ | −10.3 [−17.3, −3.3]$^{*}$ | −16.0 [−28.5, −3.1]$^{*}$ |
| | $\mathrm{ET}_{\mathrm{max}}$ | −1.6 [−3.7, +0.6] | −1.3 [−3.4, +1.0] | −0.6 [−3.7, +3.3] | −2.3 [−5.0, +0.6] | −3.8 [−8.5, +2.0] |
| | WUE | −0.6 [−4.9, +4.0] | −1.1 [−4.7, +2.9] | −3.8 [−10.5, +4.4] | −2.0 [−6.9, +3.5] | −7.6 [−15.5, +0.3] |

Table 8: Effect of adding mortality predictors on per-site prediction error (summary of Fig. 2). For every site, MAE of the model with climate, traits and mortality (C+T+M) is compared with the model without mortality (C+T), both evaluated by repeated leave-one-site-out cross-validation ($3 \times 80\%$). $\Delta$MAE = geometric mean over sites of the paired error ratio, $(\exp(\text{mean}_{\text{sites}} \ln(\text{MAE}_{\text{C+T+M}} / \text{MAE}_{\text{C+T}})) - 1) \times 100$; negative values mean that mortality reduces the error. Improved = share of sites whose MAE decreased. Sites are grouped by their maximum relative mortality across years (averaged over the five buffers, 100–500 m); High > mean + 0.5 SD (24.0%), Low < mean − 0.5 SD (10.6%), Mid in between. Stars: one-sided paired permutation test on the per-site log error ratios (100 000 random sign flips; C+T+M < C+T), Benjamini–Hochberg corrected across all 72 tests of the three learners ($^{*}\, q < 0.05$, $^{**}\, q < 0.01$, $^{***}\, q < 0.001$). Tree cover ≥ 30%, 101 sites, 361 site-years, deadtrees v5.

| Model | EFP | | All sites ($n = 101$) | Mid + High ($n = 60$) | High ($n = 22$) |
|---|---|---|---|---|---|
| XGBoost 12 months | $GPP_{sat}$ | $\Delta$MAE (%) | $-2.5$ | $+1.4$ | $+0.2$ |
| | | Improved (%) | 54 | 57 | 59 |
| | $NEP_{max}$ | $\Delta$MAE (%) | $-15.1^{**}$ | $-13.0^{**}$ | $-18.4^{*}$ |
| | | Improved (%) | 70 | 68 | 73 |
| | $ET_{max}$ | $\Delta$MAE (%) | $-1.2$ | $-3.9$ | $-0.5$ |
| | | Improved (%) | 59 | 63 | 64 |
| | WUE | $\Delta$MAE (%) | $-4.9$ | $-6.8$ | $-25.2^{*}$ |
| | | Improved (%) | 52 | 53 | 68 |
| XGBoost 24 months | $GPP_{sat}$ | $\Delta$MAE (%) | $-6.9$ | $-12.7^{*}$ | $-18.7$ |
| | | Improved (%) | 61 | 65 | 68 |
| | $NEP_{max}$ | $\Delta$MAE (%) | $-15.5^{**}$ | $-16.5^{**}$ | $-29.4^{**}$ |
| | | Improved (%) | 68 | 63 | 73 |
| | $ET_{max}$ | $\Delta$MAE (%) | $-2.0$ | $-3.1$ | $-5.0$ |
| | | Improved (%) | 54 | 63 | 68 |
| | WUE | $\Delta$MAE (%) | $-2.9$ | $-3.9$ | $-20.9$ |
| | | Improved (%) | 54 | 57 | 68 |
| LightGBM 12 months | $GPP_{sat}$ | $\Delta$MAE (%) | $-3.7$ | $-2.6$ | $-8.5$ |
| | | Improved (%) | 54 | 58 | 68 |
| | $NEP_{max}$ | $\Delta$MAE (%) | $-12.1^{**}$ | $-10.6^{*}$ | $-17.9^{*}$ |
| | | Improved (%) | 70 | 70 | 77 |
| | $ET_{max}$ | $\Delta$MAE (%) | $-4.2$ | $-5.8$ | $-1.3$ |
| | | Improved (%) | 56 | 62 | 50 |
| | WUE | $\Delta$MAE (%) | $+2.4$ | $-2.0$ | $-9.6$ |
| | | Improved (%) | 52 | 62 | 55 |
| LightGBM 24 months | $GPP_{sat}$ | $\Delta$MAE (%) | $-6.5$ | $-7.1^{*}$ | $-10.6$ |
| | | Improved (%) | 58 | 62 | 64 |
| | $NEP_{max}$ | $\Delta$MAE (%) | $-13.3^{*}$ | $-16.6^{**}$ | $-22.1^{*}$ |
| | | Improved (%) | 65 | 63 | 68 |
| | $ET_{max}$ | $\Delta$MAE (%) | $+3.7$ | $-1.2$ | $-1.1$ |
| | | Improved (%) | 56 | 58 | 59 |
| | WUE | $\Delta$MAE (%) | $-1.1$ | $+0.2$ | $-8.4$ |
| | | Improved (%) | 50 | 48 | 64 |

## A3. Direction per site: site names, 12 months and XGBoost / LightGBM

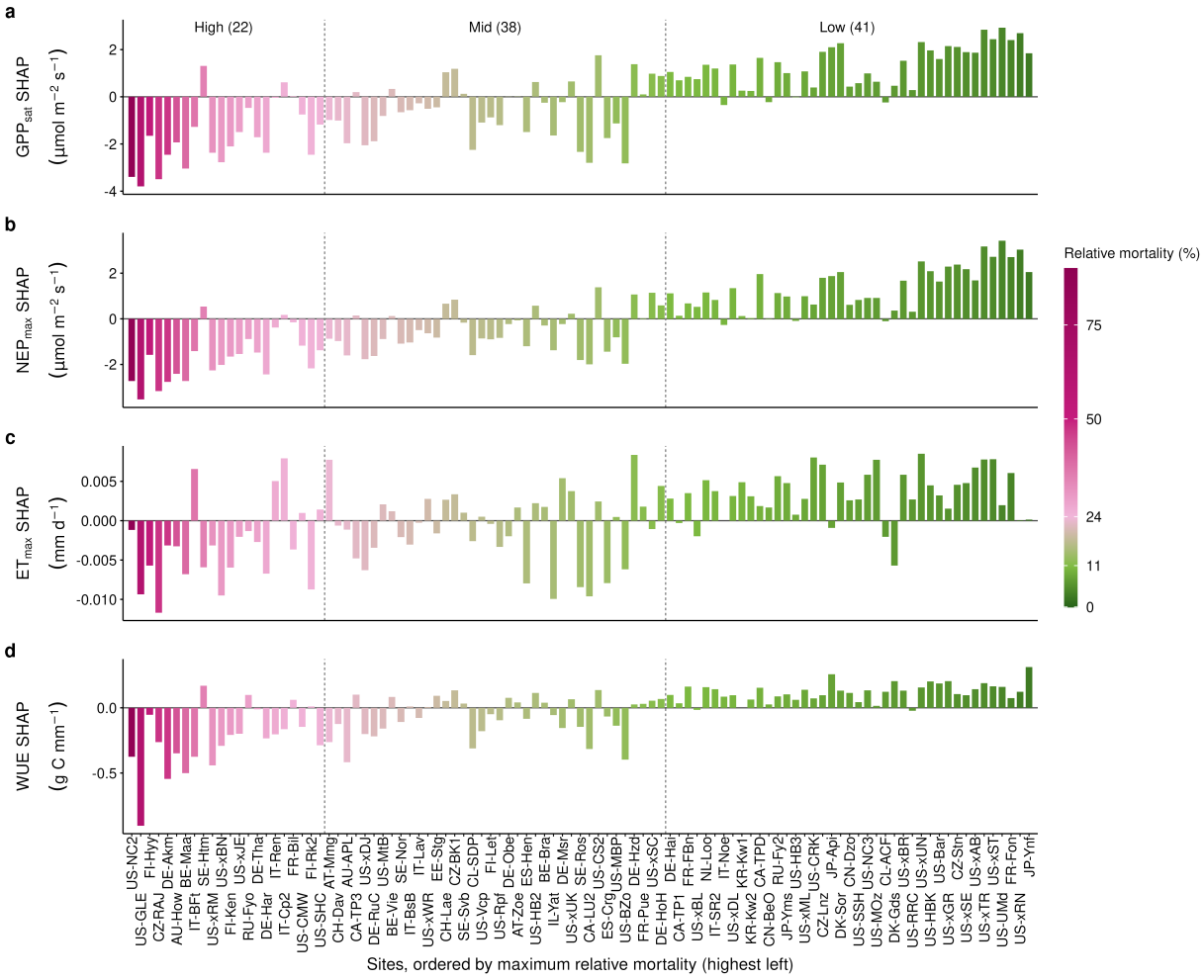



**Fig. A6** — Fig. 2, AutoGluon, 12 months.

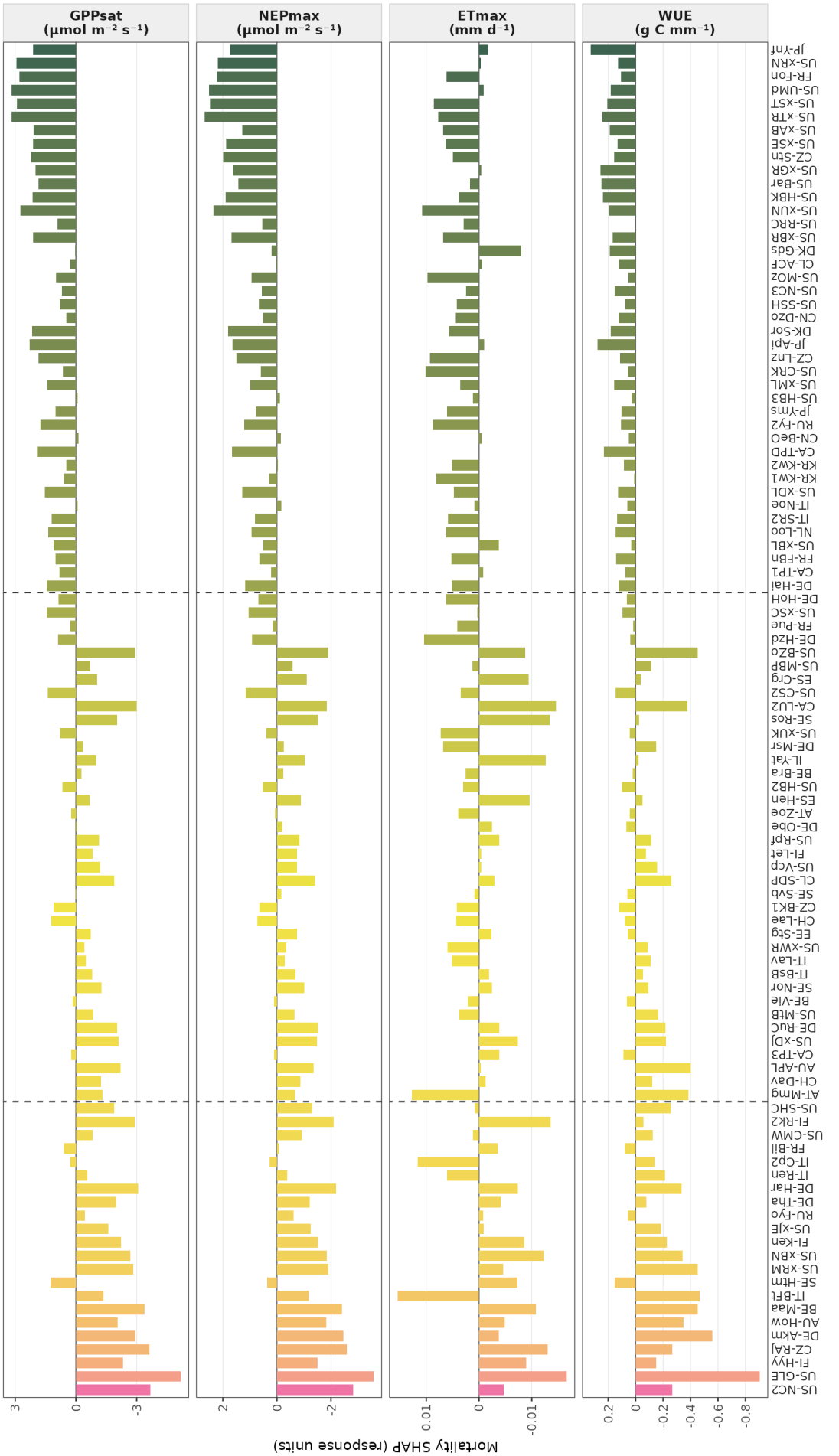



**Fig. A5** — Fig. 2 with site names, AutoGluon, 24 months.

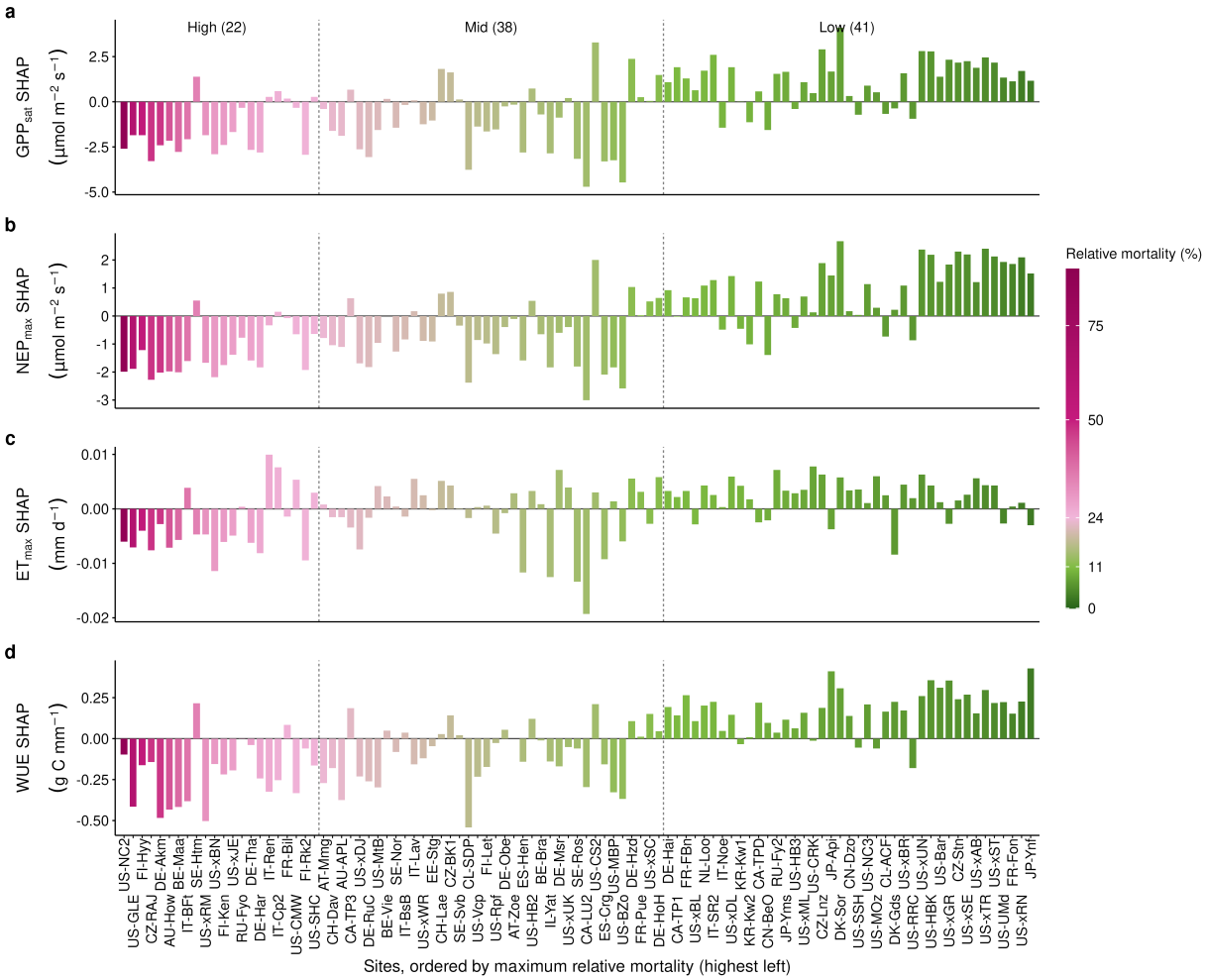



**Fig. A7** — Fig. 2, XGBoost–Optuna, 24 months.

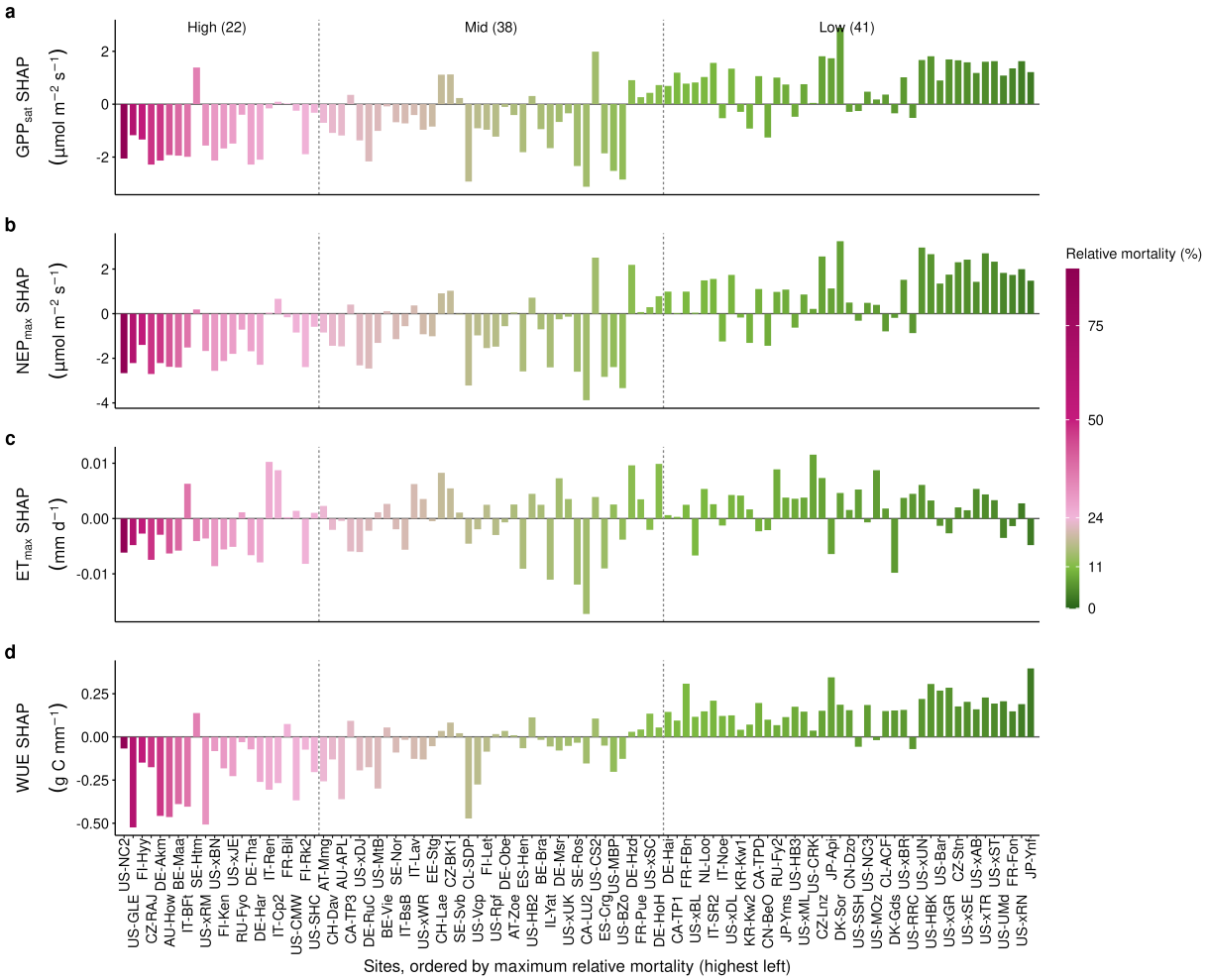



**Fig. A8** — Fig. 2, LightGBM–Optuna, 24 months.

## A4. High vs Low+Mid: all learners and relative deadwood

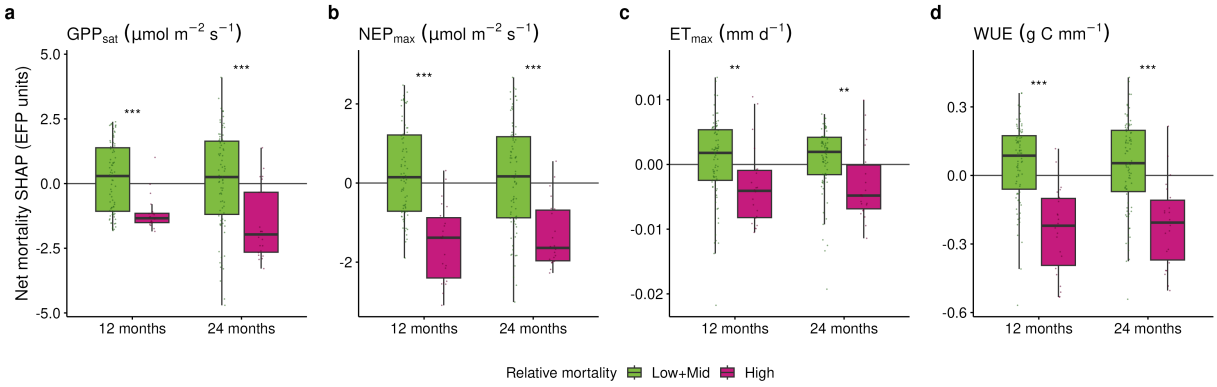



**Fig. A9** — Fig. 3, XGBoost–Optuna.

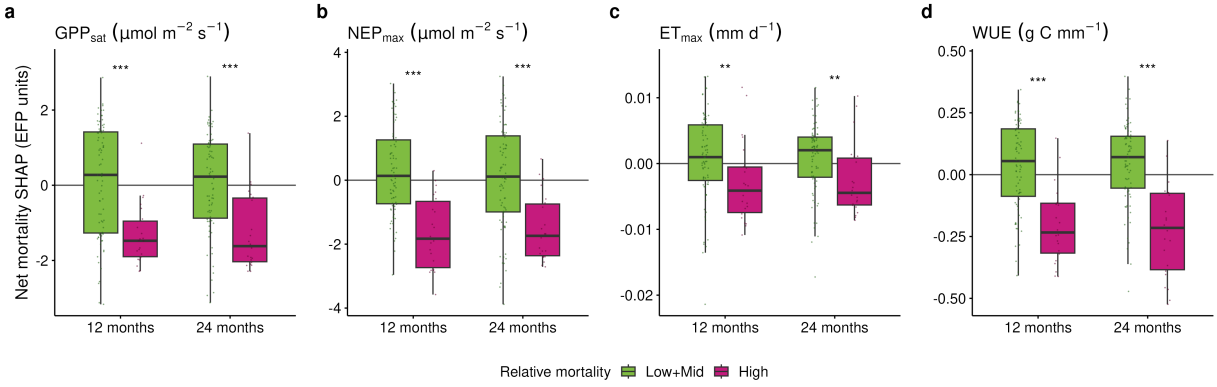



**Fig. A9b** — Fig. 3, LightGBM–Optuna.

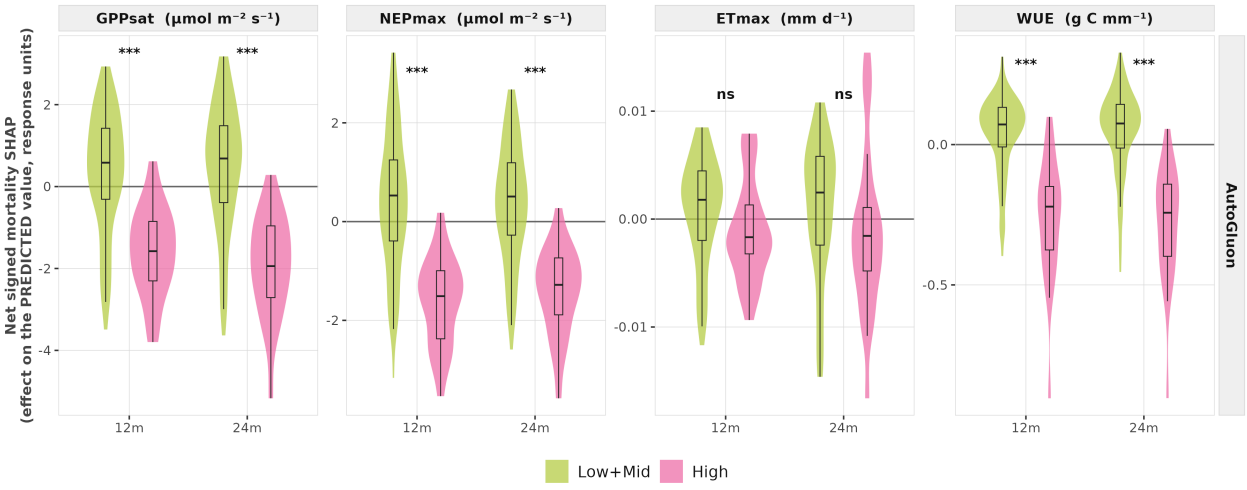



**Fig. A10** — Fig. 3 split by relative deadwood, AutoGluon: the same for GPP$_{sat}$, NEP$_{max}$ and WUE; ET$_{max}$ not significant.

## A5. Ecosystem type and tree cover

- **A** ΔRMSE by IGBP, **B** net signed mortality SHAP by IGBP, **C** tree cover vs benefit, **D** tree cover vs direction, **E** relative mortality vs direction (= Fig. 4).
- C: no relationship ($\rho$ = 0.10–0.18, n.s.). D: positive, $\rho$ = +0.66 to +0.75 (all $p < 0.001$) — at low tree cover mortality pulls predicted carbon uptake down, at high tree cover it pushes it up.

- B: DBF, EBF and MF mostly positive, CSH, OSH and WET negative, ENF slightly negative. OSH has 2 sites, WSA 1, CSH and EBF 3 each, WET 6.

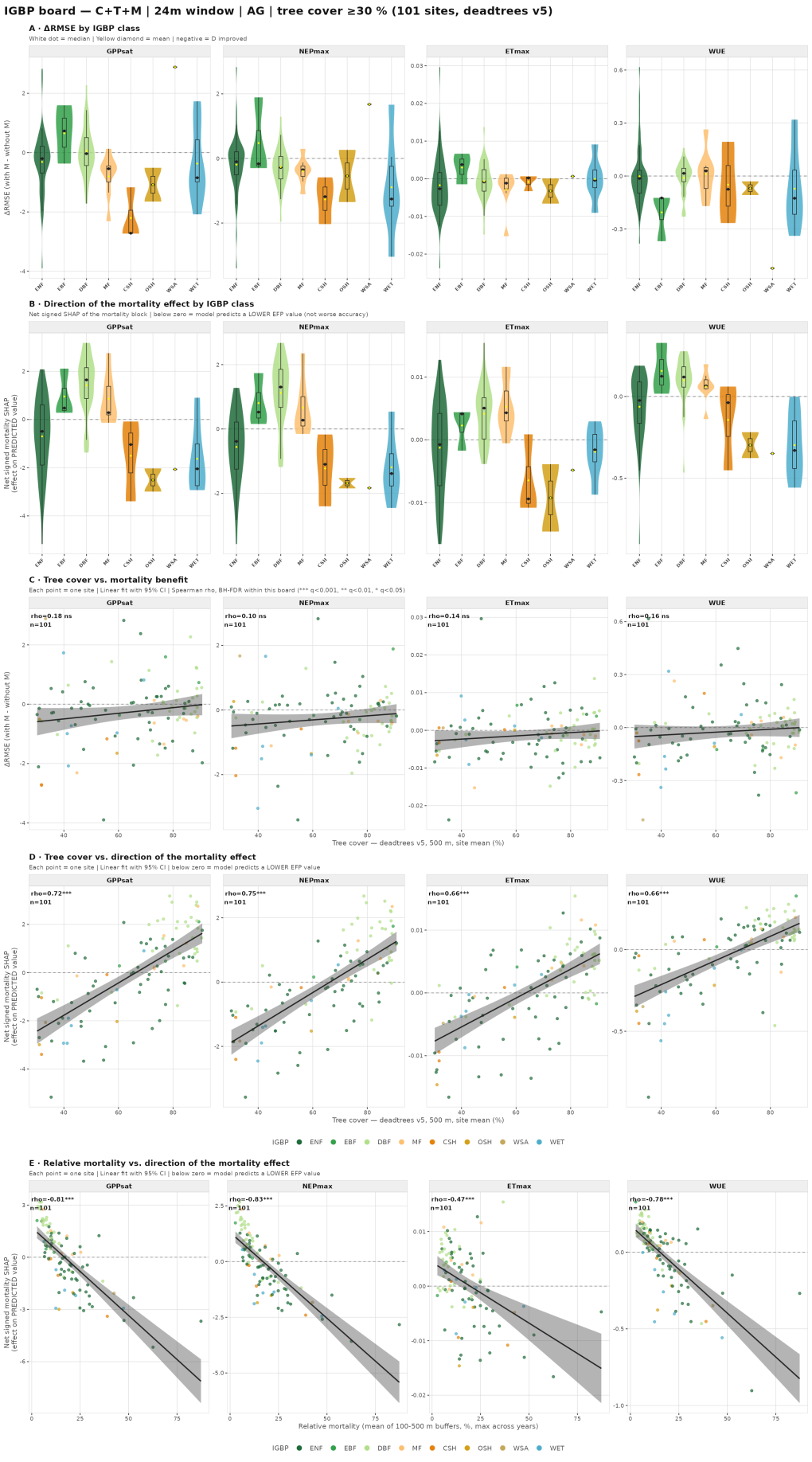



**Fig. A11** — AutoGluon, C+T+M, 24 months.

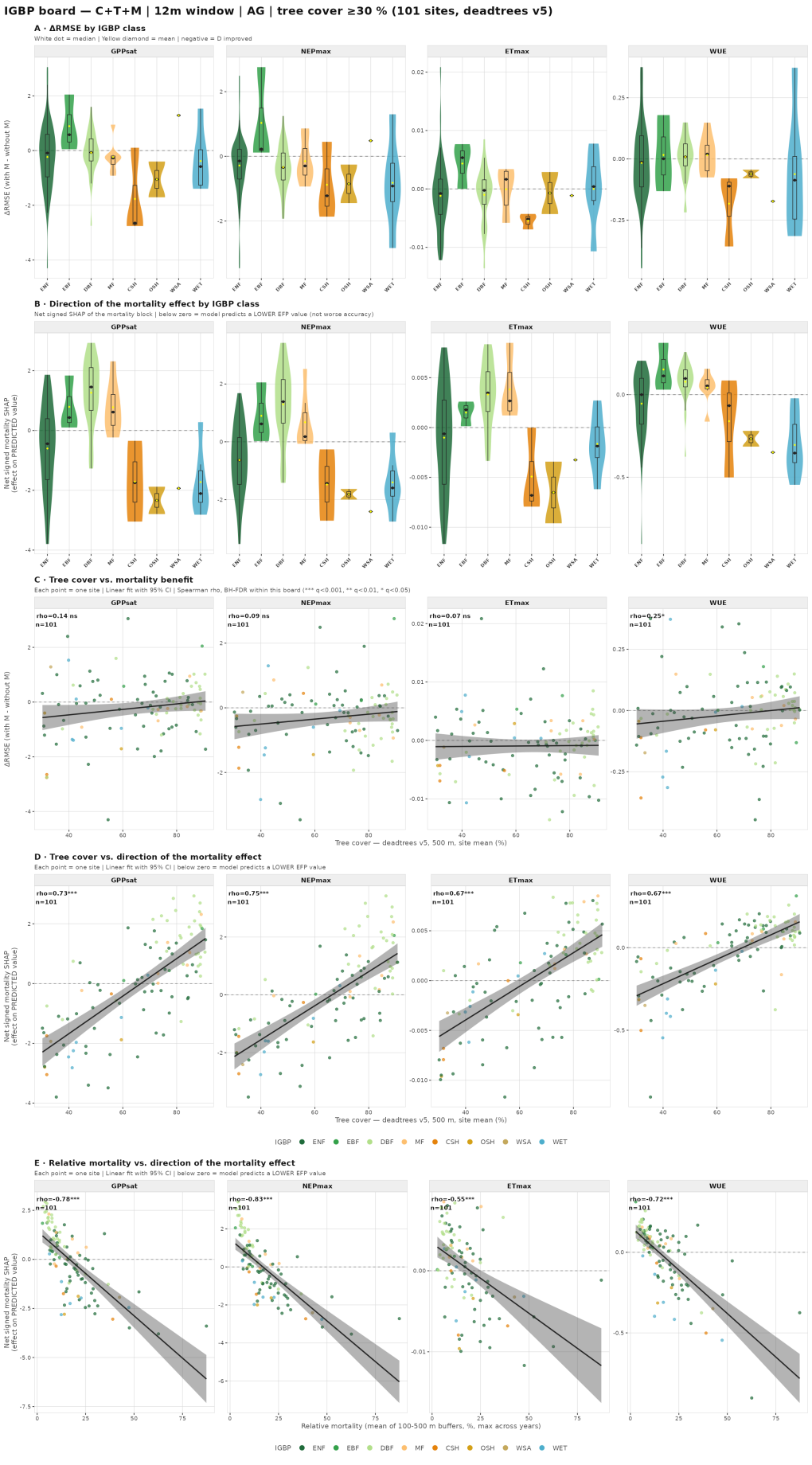



**Fig. A12** — AutoGluon, 12 months.

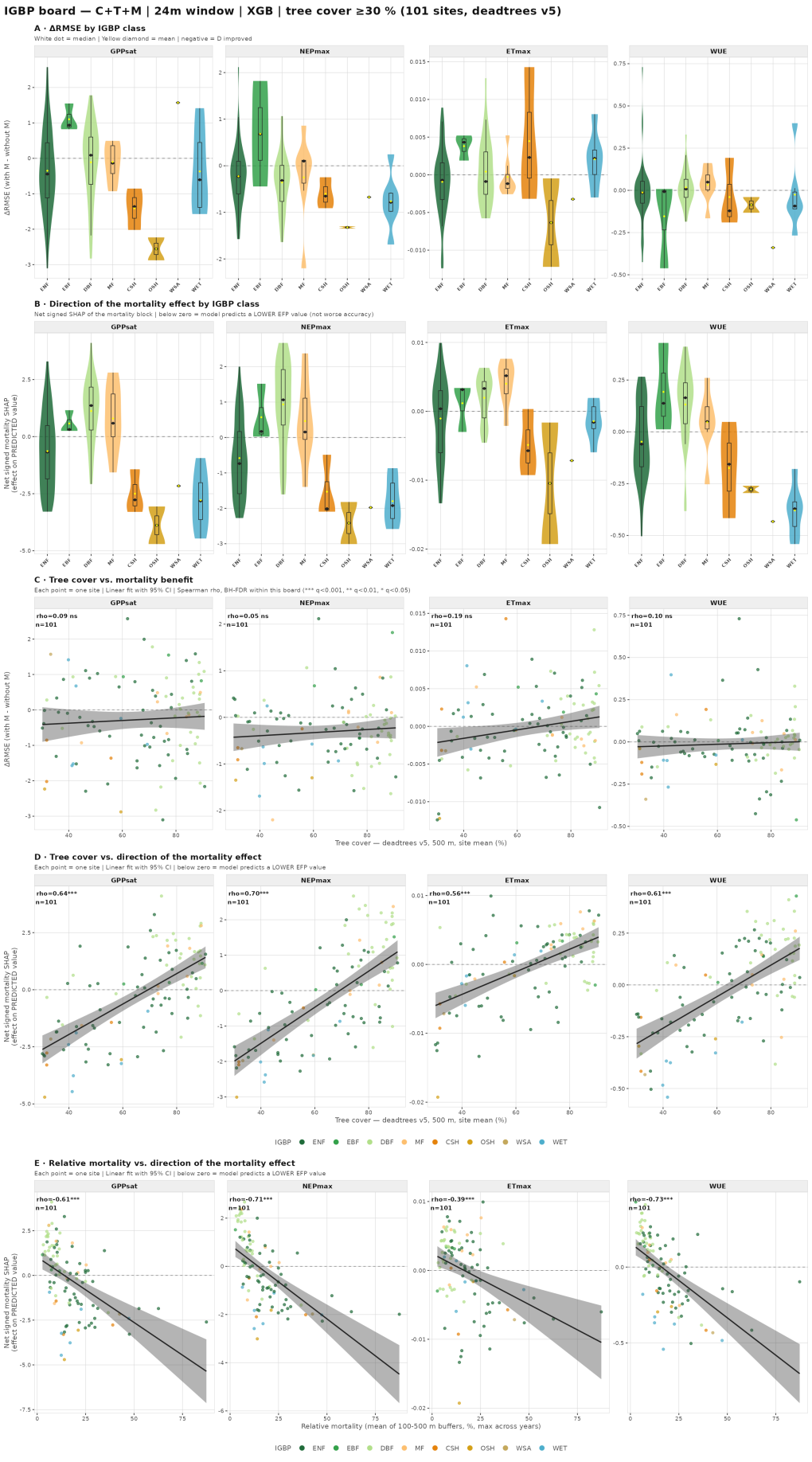



**Fig. A13** — XGBoost–Optuna, 24 months.

## A6. Mortality metrics

- Share of the mortality block's mean |SHAP| per metric, AutoGluon: relative mortality 26 %, relative deadwood 24 %, relative tree-cover loss 22 %, absolute deadwood 16 %, deadwood gain 12 %; for $ET_{max}$ tree-cover loss leads (32 %).
- Drop-column test (XGBoost and LightGBM retrained without each metric, or with only one metric added to C+T): removing any single metric changes the error by at most about 1 %, because the correlated metrics take over (relative mortality vs relative deadwood $\rho = 0.91$). Absolute deadwood gets 14–16 % of the SHAP but adds nothing when removed.

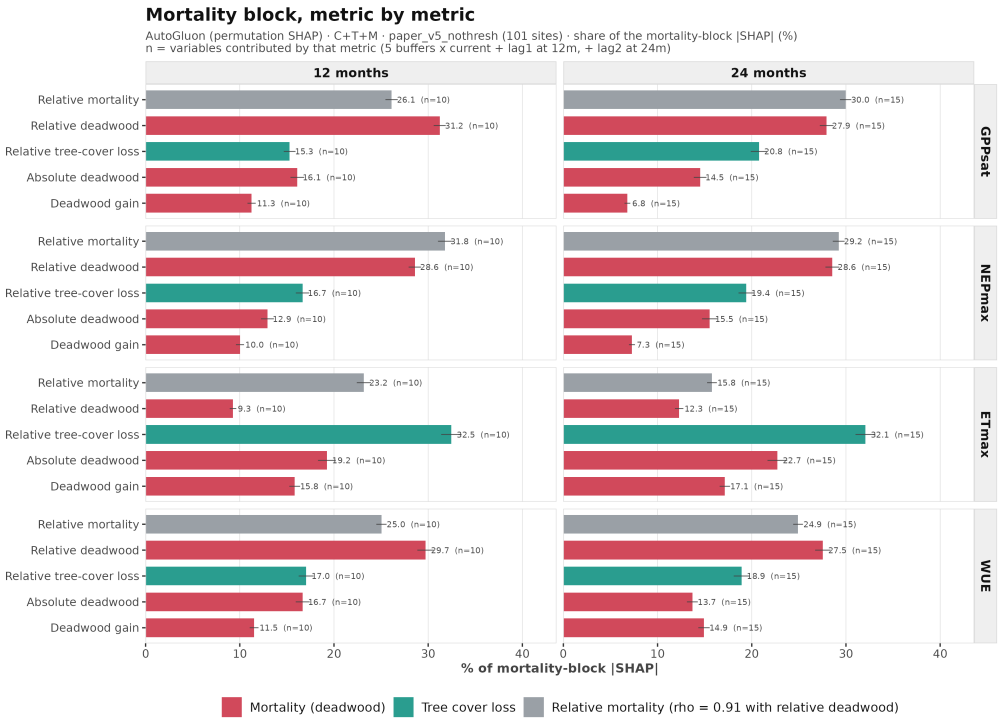



**Fig. A14** — mortality block metric by metric, AutoGluon (bars ± SE over sites).

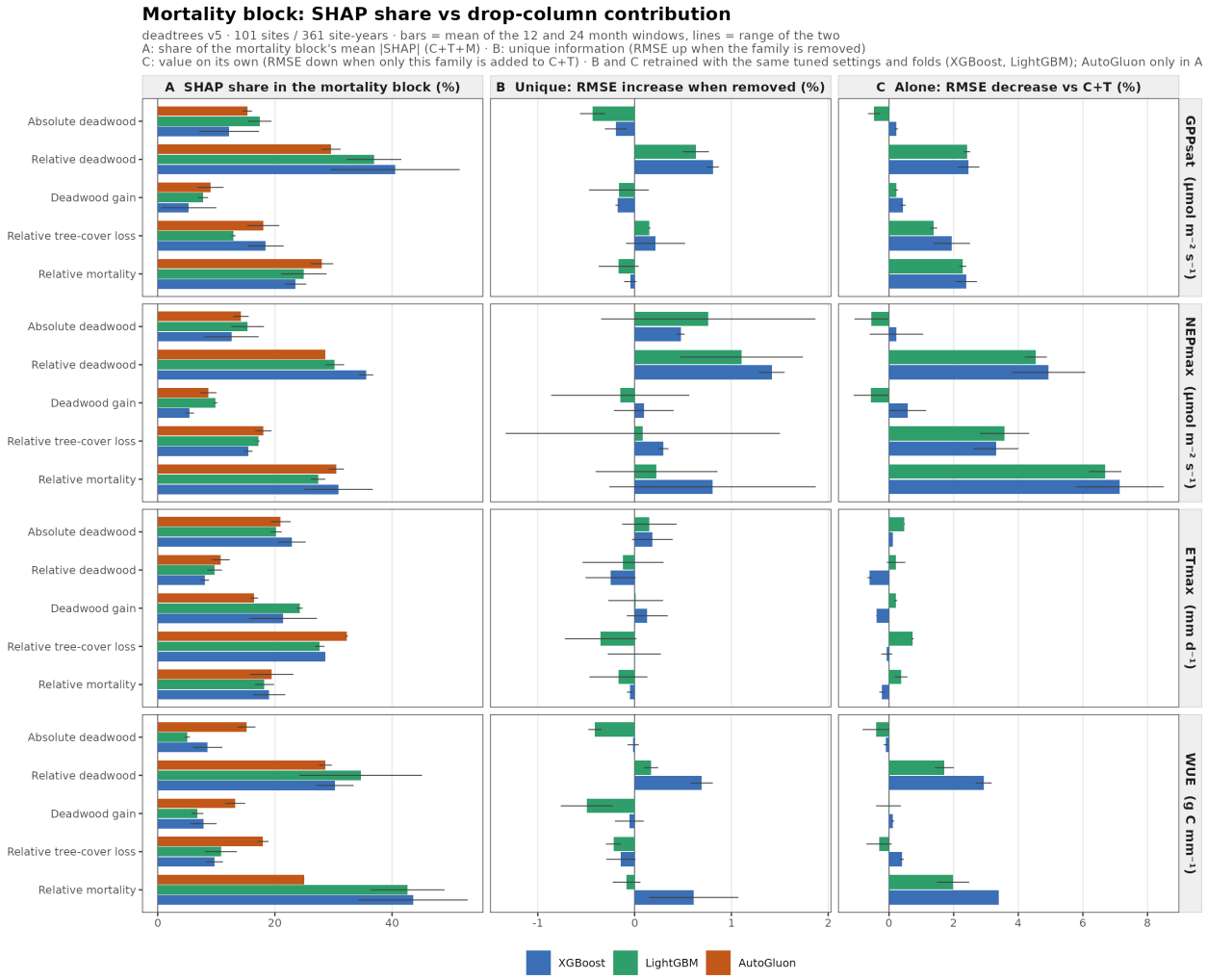



**Fig. A15** — per metric: A SHAP share in the mortality block (all learners); B RMSE increase when the metric is removed from C+T+M; C RMSE decrease when only this metric is added to C+T (B and C: XGBoost and LightGBM retrained with the same tuned settings and folds).

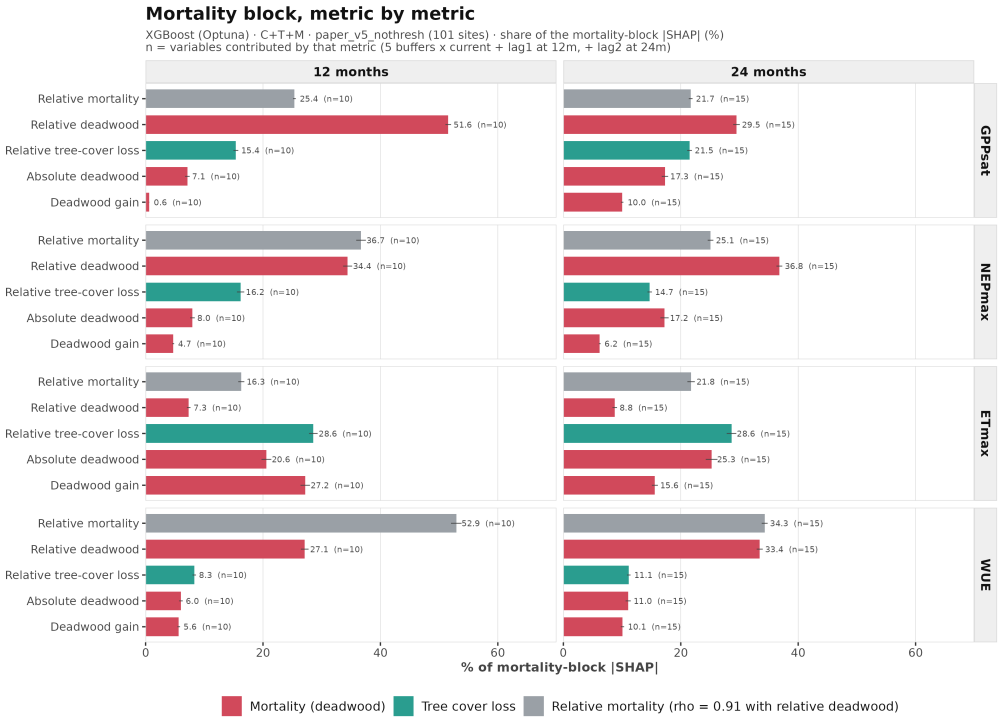


**Fig. A16** — Fig. A14 for XGBoost.

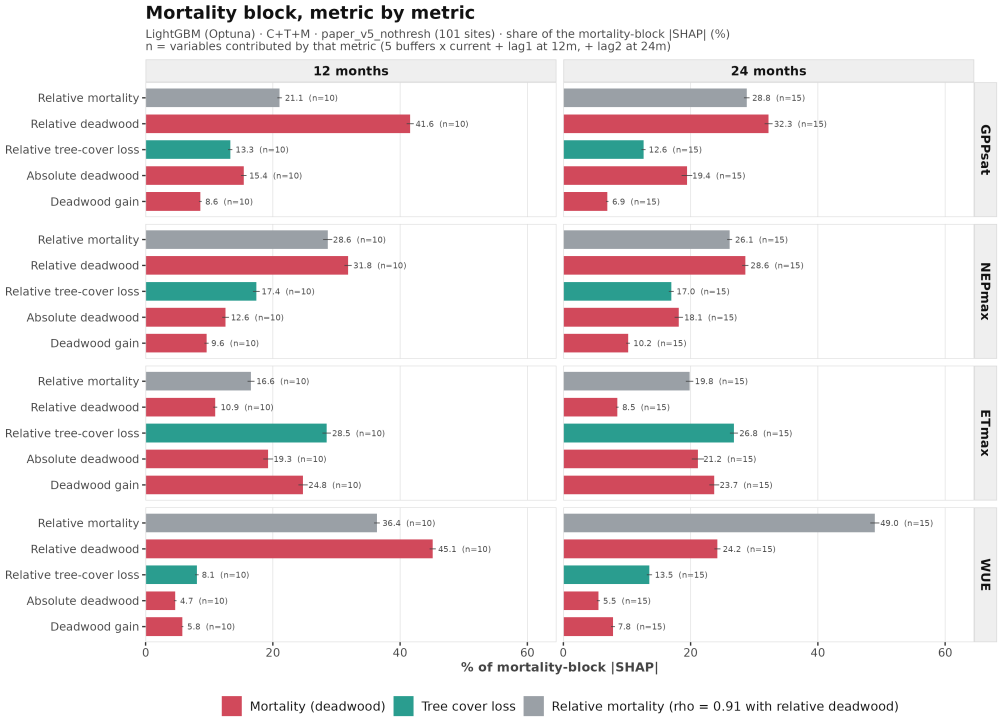



**Fig. A17** — Fig. A14 for LightGBM.

## A7. Fig. 5: other learners, 12 months, mixed models and robustness

- XGBoost and LightGBM over-predict in the >30 % class as much as AutoGluon without mortality (NEP$_{max}$ −3.6 to −4.0 $\mu$mol m$^{-2}$ s$^{-1}$); with mortality the over-prediction is smaller but stays below 0 (−1.8 to −2.4). Their correction in the >30 % class is significant for GPP$_{sat}$ (+1.4 to +1.8), NEP$_{max}$ (+1.2 to +2.2) and WUE (+0.12 to +0.19), all $q < 0.05$.
- Robustness of the mixed-model slopes (Fig. A21): mixed model, site-block bootstrap (1000 draws), leave one site out (range over 101 refits) and without the three most extreme sites (FI-Hyy, US-GLE, US-NC2). The over-prediction without mortality (middle row) holds in every check for NEP$_{max}$ and WUE; with mortality (bottom row) no check is significant for GPP$_{sat}$, NEP$_{max}$ or WUE. Whether the error reduction itself grows with mortality (top row) goes the same way but is not robust in the bootstrap.

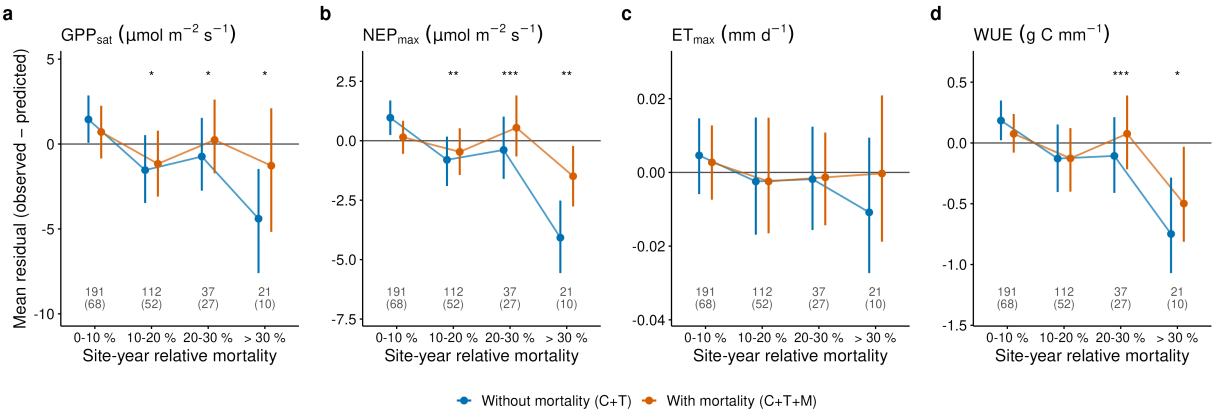



**Fig. A18** — Fig. 5 with the 12-month window, AutoGluon.

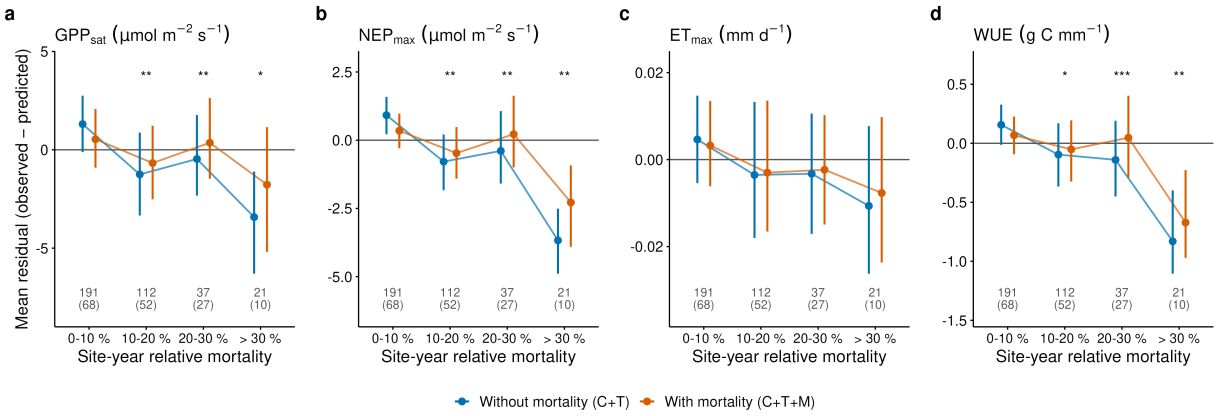



**Fig. A19** — Fig. 5, XGBoost–Optuna, 24 months.

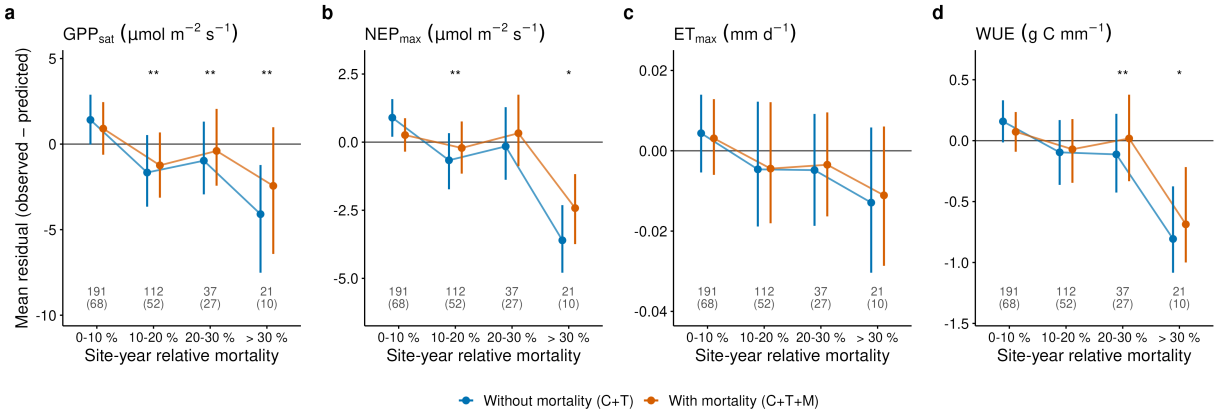



**Fig. A20** — Fig. 5, LightGBM–Optuna, 24 months.

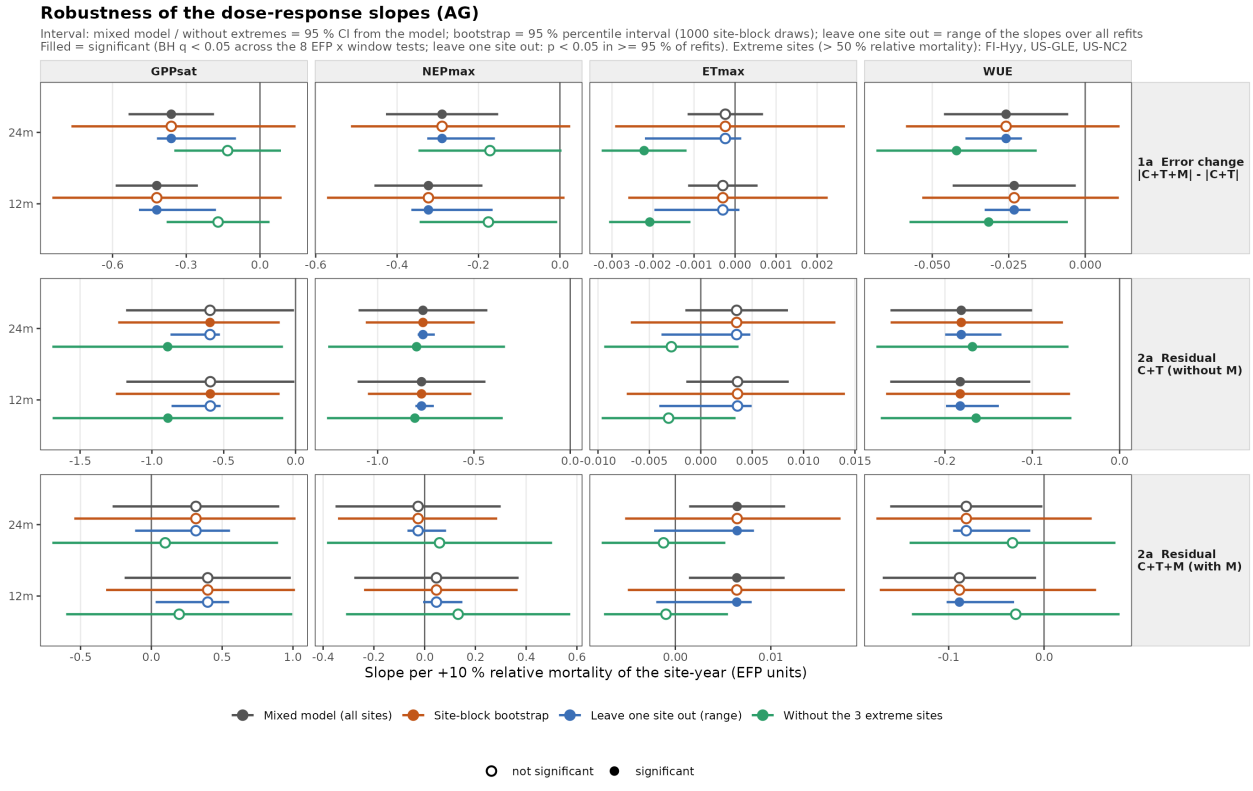



**Fig. A21** — slope per +10 % relative mortality under each robustness check, AutoGluon; filled =

significant.

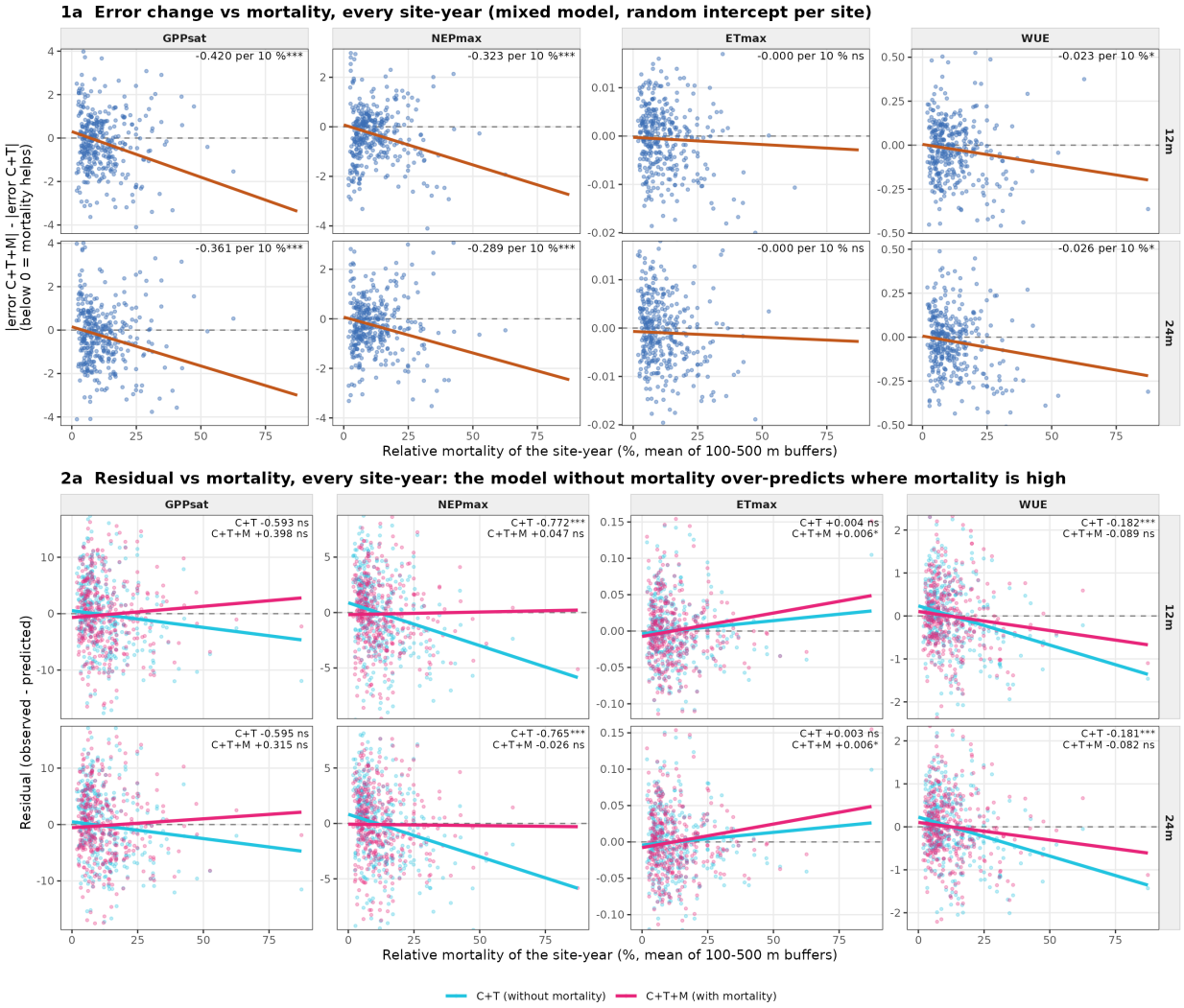



**Fig. A22** — every site-year with the mixed-model slope, AutoGluon: top, error change; bottom, residuals without (cyan) and with mortality (pink).

## A8. Circularity check: EFP-specific climate block

- Concern (Miguel): a climate variable that is part of an EFP's formula could make the prediction partly circular. The climate predictors are already strictly pre-season, but to be safe each EFP was refitted (AutoGluon, same design) **without the climate variables in its formula**: $\mathrm{GPP}_{sat}$ without radiation (light-response curve on PPFD = $\mathrm{SW}_{IN}$ × 2.11), $\mathrm{ET}_{max}$ and WUE without air temperature ($\mathrm{ET} = \mathrm{LE} / \lambda(\mathrm{T}_a)$); $\mathrm{NEP}_{max}$ has no climate variable in its formula and is unchanged. Filters ($\mathrm{SW}_{IN}$ thresholds, rain) are not part of the formulas and were kept. 36 (12 m) and 72 (24 m) climate predictors removed.

- **Skill hardly changes:** $R^2$ of C+T+M (12 / 24 m) $\mathrm{GPP}_{sat}$ 0.385 / 0.365 (before 0.367 / 0.370), $\mathrm{ET}_{max}$ 0.409 / 0.420 (0.425 / 0.433), WUE 0.281 / 0.302 (0.312 / 0.326).

- **The mortality effect is unchanged** (Table 9): WUE still improves significantly at High sites (−11 / −15 %) and in high-mortality years (−17 / −20 %); $\mathrm{GPP}_{sat}$ −4 to −16 %, significant at 12 m (all site-years −6.1 %, Mid+High years −8.3 %) but not at 24 m; $\mathrm{ET}_{max}$ not significant.

- **The over-prediction without mortality and its correction remain** (Fig. A23): in the >30 % class $\mathrm{GPP}_{sat}$ is over-predicted by 4.6 and corrected by +3.6 $\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ ($q = 0.04$; 12 m:

+3.7, $q = 0.009$), WUE by 0.71 and corrected by $+0.25\,\mathrm{g\,C\,mm^{-1}}$ ($q = 0.04$; 12 m $q = 0.002$); $\mathrm{NEP}_{max}$ as before.

- **So:** the mortality results do not depend on climate variables that enter the EFP formulas — no circularity from the climate block.

Table 9: Paired bootstrap of the change in MAE when mortality is added (C+T → C+T+M), AutoGluon and AutoGluon, EFP-specific climate. Cells: % change of the error, $100\,(\mathrm{MAE_{C+T+M}} / \mathrm{MAE_{C+T}} - 1)$, with the 95 % bootstrap interval in brackets. *Site-year level*: every site-year is an observation (pooled MAE; no averaging over years). Years of one site are not independent, so whole sites are resampled (10 000 bootstrap draws) and the C+T / C+T+M labels are swapped per site for the paired permutation test (10 000 draws, one-sided; stars, Benjamini–Hochberg corrected across both runs: $^{*}q < 0.05$, $^{**}q < 0.01$, $^{***}q < 0.001$). High sites: site maximum of relative mortality > 24.0 %; the last two columns keep only the site-years whose own relative mortality exceeds 10.6 % (Mid + High) or 24.0 % (High). Column heads: sites / site-years. 101 sites, 361 site-years, deadtrees v5.

| | | All sites (101 / 361) | High sites (22 / 80) | Mid + High years (60 / 160) | High years (22 / 40) |
|---|---|---|---|---|---|
| AutoGluon 12 months | $\mathrm{GPP_{sat}}$ | −4.1 [−8.0, −0.2]* | −7.2 [−18.1, +2.9] | −6.7 [−12.8, −0.7]* | −12.0 [−29.5, +4.9] |
| | $\mathrm{NEP_{max}}$ | −11.8 [−17.0, −6.3]** | −22.6 [−33.6, −11.1]** | −12.5 [−20.1, −4.9]** | −31.6 [−43.2, −16.0]* |
| | $\mathrm{ET_{max}}$ | −1.3 [−4.2, +1.3] | −2.9 [−8.4, +2.1] | −3.9 [−7.7, +0.0]* | −9.7 [−20.6, +2.7] |
| | WUE | −4.3 [−8.1, −0.2]* | −10.9 [−17.8, −4.3]** | −7.2 [−12.2, −1.8]* | −16.9 [−25.2, −5.1]* |
| AutoGluon 24 months | $\mathrm{GPP_{sat}}$ | −5.7 [−9.5, −1.7]* | −7.1 [−17.2, +3.6] | −8.3 [−14.2, −2.6]* | −11.1 [−28.0, +8.1] |
| | $\mathrm{NEP_{max}}$ | −10.8 [−16.0, −5.1]** | −17.5 [−30.7, −2.1]* | −11.6 [−19.8, −2.7]* | −29.2 [−42.9, −8.6]* |
| | $\mathrm{ET_{max}}$ | −2.0 [−5.0, +0.9] | −5.3 [−13.1, +1.2] | −5.4 [−9.6, −1.3]* | −11.4 [−23.7, +3.2] |
| | WUE | −4.7 [−9.1, −0.1]* | −14.8 [−23.9, −6.4]* | −8.2 [−14.5, −1.5]* | −21.8 [−34.5, −8.2]* |
| AutoGluon, EFP-specific climate 12 months | $\mathrm{GPP_{sat}}$ | −6.1 [−10.5, −1.7]* | −12.1 [−24.4, +0.5] | −8.3 [−15.6, −1.1]* | −16.1 [−35.0, +5.0] |
| | $\mathrm{NEP_{max}}$ | −11.8 [−17.0, −6.3]** | −22.6 [−33.6, −11.1]** | −12.5 [−20.1, −4.9]** | −31.6 [−43.2, −16.0]* |
| | $\mathrm{ET_{max}}$ | −1.5 [−4.4, +1.4] | +0.4 [−5.5, +8.0] | −2.2 [−5.9, +2.4] | −2.0 [−17.0, +14.8] |
| | WUE | −4.3 [−7.9, −0.1]* | −11.2 [−18.6, −4.4]* | −6.3 [−12.0, −0.2]* | −16.8 [−26.6, −5.5]* |
| AutoGluon, EFP-specific climate 24 months | $\mathrm{GPP_{sat}}$ | −4.1 [−8.4, +0.2] | −4.5 [−19.0, +11.6] | −4.7 [−12.6, +3.2] | −9.9 [−33.3, +19.7] |
| | $\mathrm{NEP_{max}}$ | −10.8 [−16.0, −5.1]** | −17.5 [−30.7, −2.1]* | −11.6 [−19.8, −2.7]* | −29.2 [−42.9, −8.6]* |
| | $\mathrm{ET_{max}}$ | −1.1 [−4.1, +1.7] | −1.1 [−8.3, +6.1] | −3.5 [−8.0, +1.1] | −6.3 [−20.7, +9.3] |
| | WUE | −4.2 [−8.5, +0.5] | −14.5 [−22.5, −6.6]** | −7.9 [−14.0, −1.2]* | −20.4 [−29.2, −9.5]* |

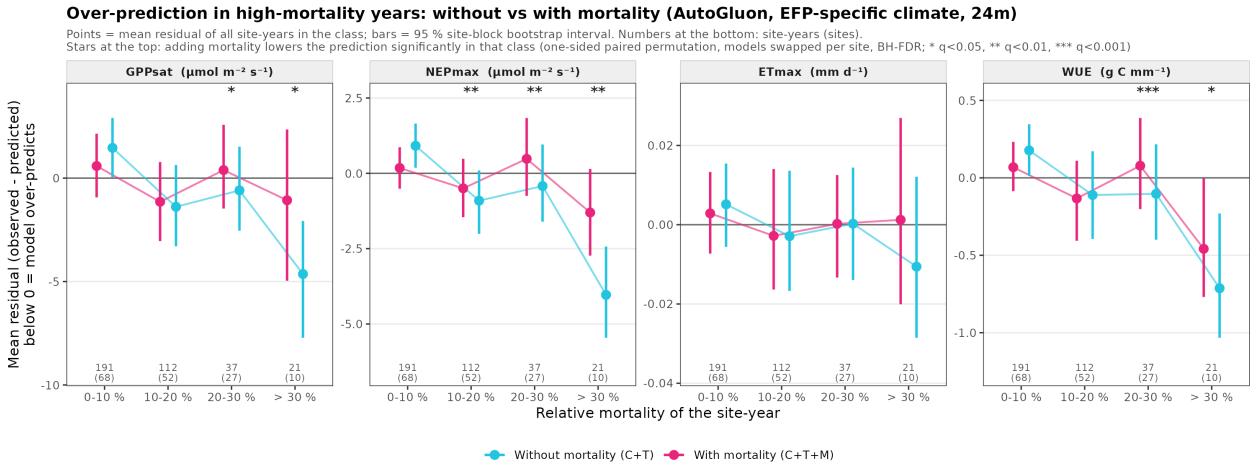



**Fig. A23** — as Fig. 5, AutoGluon with the EFP-specific climate block, 24 months.